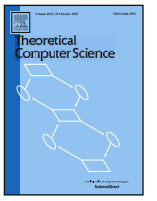
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On boolean hyperdoctrines and peircean bicategories

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ABSTRACT

Fo-bicategories can be seen as a categorical version of Peirce's calculus of relations. What makes them particularly interesting is that their rules form a complete system for first-order logic using only equations–no extra logical machinery is needed. In this paper, we show how fo-bicategories relate to Lawvere's hyperdoctrines. To make our main result easier to prove and understand, we introduce a simpler structure called peircean bicategories, which capture the essence of fo-bicategories in a more straightforward way.

1. Introduction

The first appearances of the characteristic features of first-order logic can be traced back to the works of Peirce [1] and Frege [2]. Frege was mainly motivated by the pursuit of a rigorous foundation for mathematics: his work was inspired by real analysis, bringing the concept of functions and variables into the logical realm [3]. On the other hand Peirce, inspired by the work of De Morgan [4] on relational reasoning, introduced a calculus in which operations allow the combination of relations and adhere to a set of algebraic laws. Like Boole's algebra of classes [5], Peirce's calculus of relations does not feature variables nor quantifiers and its sole deduction rule is substituting equals by equals.

Despite several negative results regarding axiomatisations for the entire calculus [6] and various fragments thereof [7–11], its lack of binder-related complexities, coupled with purely equational proofs, has rendered the calculus of relations highly influential in computer science, e.g., in the context of database theory [12], programming languages [13–17] and proof assistants [18–20].

In logic, the calculus played a secondary role for many years, likely because it is strictly less expressive than first-order logic [21]. This was until Tarski in [22] recognized its algebraic flavour and initiated a program of algebraizing first-order logic, including works such as [23–25]. Quoting Quine [25]:

> "Logic in his adolescent phase was algebraic. There was Boole's algebra of classes and Peirce's algebra of relations. But in 1879 logic come of age, with Frege's quantification theory. Here the bound variables, so characteristic of analysis rather than of algebra, became central to logic."

Such a perspective, which regarded algebraic aspects and those concerning quantifiers as separate entities, changed with the work of Lawvere.

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Thanks to the recent development of a new branch of mathematics, namely category theory, Lawvere introduced in [26–28] *hyperdoctrines* which enabled the study of logic from a pure algebraic perspective. The crucial insight of Lawvere was to show that quantifiers, as well as many logical constructs, can be algebraically captured through the crucial notion of adjointness. Hyperdoctrines, along with many categorical structures related to logics, such as regular, Heyting, and boolean categories [29,30], align with Frege's functional perspective: arrows represent functions (terms), and relations are derived through specific constructions.

In the last decade, the paradigm shift towards treating data as a physical resource has motivated many computer scientists to move from traditional term-based (cartesian) syntax toward a string diagrammatic (monoidal) syntax [31,32] (see e.g., [33–42]). This shift in syntax enables an extension of Peirce's calculus of relations that is as expressive as first-order logic, accompanied by an axiomatisation that is purely equational and complete. The axioms are those of *first-order bicategories* [43]: see Figs. 2–5. In essence, a first-order bicategory, shortly fo-bicategory, encompasses a cartesian and a cocartesian bicategory [44], interacting as a linear bicategory [45], while additionally satisfying linear versions of Frobenius equations and adjointness conditions.

In this paper, we reconcile Lawvere's understanding of logic with Peirce's calculus of relations by illustrating a formal correspondence between boolean hyperdoctrines and first-order bicategories.

To reach such a correspondence, we found convenient to introduce the novel notion of *peircean bicategories*: these are cartesian bicategories with each homset carrying a boolean algebra where the negation behaves appropriately with *maps* – special arrows that intuitively generalize functions. Our first result (Theorem 3.1) establishes that peircean bicategories are equivalent to first-order bicategories.

While the definition of peircean bicategories is not purely equational, as in the case of fo-bicategories, it is notably more concise. Moreover, it allows us to reuse from [46] an adjunction between cartesian bicategories and *elementary and existential doctrines* [47–49], which are a generalisation of hyperdoctrines, corresponding to the regular (i.e., $\exists, =, \top, \wedge$) fragment of first-order logic.

Our main result (Theorem 5.1) reveals an adjunction between the category of first-order bicategories and the category of boolean hyperdoctrines. One can perceive this as a logical analogue of Fox's theorem [50], which establishes an equivalence between categories with finite products and monoidal categories equipped with natural comonoids. The latter notion, like the one of fo-bicategories, is purely equational.

It is essential to note that our theorem establishes an adjunction rather than an equivalence. The discrepancy can be intuitively explained by observing that, akin to first-order logic, terms and formulas are distinct entities in hyperdoctrines. This differentiation does not exist in the calculus of relations or first-order bicategories. For instance, given two terms t_1 and t_2, the hyperdoctrine where the formula $t_1 = t_2$ is true differs from the hyperdoctrine where t_1 and t_2 are equated as terms, a distinction not present in fo-bicategories. These issues, related to the extensionality of equality, are thoroughly analyzed in the context of doctrines in [48] and, more generally, in that of fibrations in [51].

Leveraging another result from [46], we demonstrate (Theorem 5.2) that the adjunction in Theorem 5.1 becomes an equivalence when restricted to well-behaved hyperdoctrines (i.e., those whose equality is extensional and satisfying the rule of unique choice [52]).

Finally, we show (Corollary 6.4) that *boolean categories* [30] are equivalent to those first-order bicategories that are *functionally complete*– a concept originally introduced in [44] within the setting of cartesian bicategories. The core of our argument lies in demonstrating that a cartesian bicategory is functionally complete if and only if the corresponding doctrine has *full comprehensions* (Theorem 6.2). By combining this result with a theorem from [52], we arrive at the desired equivalence.

Synopsis: In Section 2, we provide a review of (co)cartesian bicategories, linear bicategories, and fo-bicategories. Section 3 introduces peircean bicategories and establishes their equivalence with fo-bicategories. Section 4 covers a recap of elementary and existential doctrines and recall the adjunction and the equivalence from [46]. Section 5 first recalls boolean hyperdoctrines and then illustrates the correspondences with first-order bicategories. Section 6 elucidates the correspondence with boolean categories. The appendix contains the proof of minor results not illustrated in the main text.

Contributions: This paper is an extended version of [53]. In addition to full proofs and further examples, it includes new material on the correspondence with boolean categories, presented in Section 6. The main results are summarised in Fig. 1, where the first two rows report the isomorphism between fo-bicategories ($\mathbb{FOB}$) and peircean bicategories ($\mathbb{PB}$), as well as the adjunction with boolean hyperdoctrines ($\mathbb{BHD}$) and the equivalence with boolean hyperdoctrines with extensional equality and satisfying the rule of unique choice ($\overline{\mathbb{BHD}}$); the last row contains the additional result: namely, the equivalence between functionally complete fo-bicategories and boolean hyperdoctrines with full comprehensions ($\overline{\mathbb{BHD}}_c$), that allows us to conclude the equivalence of the former with boolean categories ($\mathbb{BC}$).

Terminology and Notation: All bicategories considered in this paper are just poset-enriched symmetric monoidal categories. In these bicategories, given $c : X \to Y$ and $d : Y \to X$, we say that c is left adjoint to d, or d is right adjoint to c, if $id_X^\circ \leq c \fatsemi d$ and $d \fatsemi c \leq id_Y^\circ$. For a bicategory $\mathbf{C}$, we will write $\mathbf{C}^{\mathrm{op}}$ for the bicategory having the same objects as $\mathbf{C}$ but homsets $\mathbf{C}^{\mathrm{op}}[X, Y] \stackrel{\text{def}}{=} \mathbf{C}[Y, X]$. Similarly, we will write $\mathbf{C}^{\mathrm{co}}$ to denote the bicategory having the same objects and arrows of $\mathbf{C}$ but equipped with the reversed ordering $\geq$. The cartesian bicategories in this paper are called in [44] cartesian bicategories of relations. We refer the reader to [43, Rem. 2] for a comparison with the presentation of linear bicategories in [45].

2. From cartesian to first-order bicategories

In this section we recall the notion of *first-order bicategory* from [43]. To provide a preliminary intuition, it is convenient to consider **Rel**, the first-order bicategory of sets and relations.

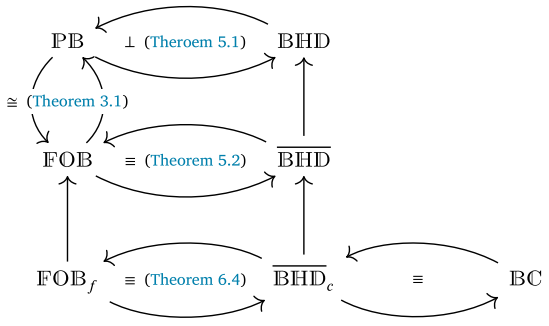


Fig. 1. A summary of the main results.

It is well known that sets and relations form a symmetric monoidal category, hereafter denoted as **Rel**°, with composition, identities, monoidal product and symmetries defined as

$$\begin{array}{rcl} a \fatsemi b & \stackrel{\text{def}}{=} & \{(x,z) \mid \exists y \in Y \,.\, (x,y) \in a \wedge (y,z) \in b\} \subseteq X \times Z \\ id^{\circ}_{X} & \stackrel{\text{def}}{=} & \{(x,y) \mid x = y\} \subseteq X \times X \\ a \otimes c & \stackrel{\text{def}}{=} & \{((x,z),(y,v)) \mid (x,y) \in a \wedge (z,v) \in c\} \subseteq (X \times Z) \times (Y \times V) \\ \sigma^{\circ}_{X,Y} & \stackrel{\text{def}}{=} & \{((x,y),(y',x')) \mid x = x' \wedge\ y = y'\} \subseteq (X \times Y) \times (Y \times X) \end{array} \tag{1}$$

for all sets X, Y, Z, V and relations $a \subseteq X \times Y$, $b \subseteq Y \times Z$ and $c \subseteq Z \times V$.

As originally observed by Peirce in [54], beyond $\fatsemi$ there exists another form of relational composition that enjoys noteworthy algebraic properties. This different composition gives rise to another symmetric monoidal category of sets and relations, hereafter denoted by **Rel**$^{\bullet}$ and defined as follows.

$$\begin{array}{rcl} a \semi b & \stackrel{\text{def}}{=} & \{(x,z) \mid \forall y \in Y \,.\, (x,y) \in a \vee (y,z) \in b\} \subseteq X \times Z \\ id^{\bullet}_{X} & \stackrel{\text{def}}{=} & \{(x,y) \mid x \neq y\} \subseteq X \times X \\ a \obar c & \stackrel{\text{def}}{=} & \{((x,z),(y,v)) \mid (x,y) \in a \vee (z,v) \in c\} \subseteq (X \times Z) \times (Y \times V) \\ \sigma^{\bullet}_{X,Y} & \stackrel{\text{def}}{=} & \{((x,y),(y',x')) \mid x \neq x' \vee\ y \neq y'\} \subseteq (X \times Y) \times (Y \times X) \end{array} \tag{2}$$

Note that $\otimes$ and $\obar$ are both defined on objects as the cartesian product of sets and have as unit the singleton set $I \stackrel{\text{def}}{=} \{\star\}$. Both **Rel**° and **Rel**$^{\bullet}$ are poset-enriched symmetric monoidal categories when taking as ordering the inclusion $\subseteq$.

As we will explain in Section 2.1, the relations defined for all sets X as

$$\begin{array}{rcl|rcl} \blacktriangleleft^{\circ}_{X} & \stackrel{\text{def}}{=} & \{(x,\ (y,z)) \mid x = y \wedge x = z\} \subseteq X \times (X \times X) & !^{\circ}_{X} & \stackrel{\text{def}}{=} & \{(x,\star) \mid x \in X\} \subseteq X \times I \\ \blacktriangleright^{\circ}_{X} & \stackrel{\text{def}}{=} & \{((y,z),\ x) \mid x = y \wedge x = z\} \subseteq (X \times X) \times X & ¡^{\circ}_{X} & \stackrel{\text{def}}{=} & \{(\star,x) \mid x \in X\} \subseteq I \times X \\ \hline \blacktriangleleft^{\bullet}_{X} & \stackrel{\text{def}}{=} & \{(x,\ (y,z)) \mid x \neq y \vee x \neq z\} \subseteq X \times (X \times X) & !^{\bullet}_{X} & \stackrel{\text{def}}{=} & \varnothing \subseteq X \times I \\ \blacktriangleright^{\bullet}_{X} & \stackrel{\text{def}}{=} & \{((y,z),\ x) \mid x \neq y \vee x \neq z\} \subseteq (X \times X) \times X & ¡^{\bullet}_{X} & \stackrel{\text{def}}{=} & \varnothing \subseteq I \times X \end{array} \tag{3}$$

make **Rel**° a cartesian bicategory, while **Rel**$^{\bullet}$ a cocartesian one.

Intuitively, a first-order bicategory **C** consists of a cartesian bicategory **C**°, called the "white structure", and a cocartesian bicategory **C**$^{\bullet}$, called the "black structure", that interact by obeying the same laws of **Rel**° and **Rel**$^{\bullet}$. The name "first-order" is due to the fact that such laws provide a complete system of axioms for first-order logic.

The axioms can be conveniently given by means of a graphical representation inspired by string diagrams [31,32]: composition is depicted as horizontal composition while the monoidal product by vertically "stacking" diagrams. However, since there are two compositions $\fatsemi$ and $\semi$ and two monoidal products $\otimes$ and $\obar$, to distinguish them we use different colors. All white constants have white background, mutatis mutandis for the black ones: for instance $\blacktriangleleft^{\circ}_{X}$ and $\blacktriangleright^{\bullet}_{X}$ are drawn  and  , while for some arrows a, b, c, d of the appropriate type, $(a \otimes c) \semi (b \obar d)$ is drawn as on the right of (ν°_{l}) in Fig. 4.

2.1. Cartesian bicategories

We commence our technical exposition with the notion of cartesian bicategories by Carboni and Walters [44].

Definition 2.1. A *cartesian bicategory* $(\mathbf{C}, \otimes, I, \blacktriangleleft^{\circ}, !^{\circ}, \blacktriangleright^{\circ}, ¡^{\circ})$, shorthand $(\mathbf{C}, \blacktriangleleft^{\circ}, \blacktriangleright^{\circ})$, is a poset-enriched symmetric monoidal category $(\mathbf{C}, \otimes, I)$ and, for every object X in **C**, arrows $\blacktriangleleft^{\circ}_{X} : X \to X \otimes X$, $!^{\circ}_{X} : X \to I$, $\blacktriangleright^{\circ}_{X} : X \otimes X \to X$, $¡^{\circ}_{X} : I \to X$ such that

1. $(\blacktriangleleft^{\circ}_{X}, !^{\circ}_{X})$ is a *comonoid* and $(\blacktriangleright^{\circ}_{X}, ¡^{\circ}_{X})$ a *monoid*, i.e., the equalities ($\blacktriangleleft^{\circ}$-as), ($\blacktriangleleft^{\circ}$-un), ($\blacktriangleleft^{\circ}$-co) and ($\blacktriangleright^{\circ}$-as), ($\blacktriangleright^{\circ}$-un), ($\blacktriangleright^{\circ}$-co) in Fig. 2 hold;

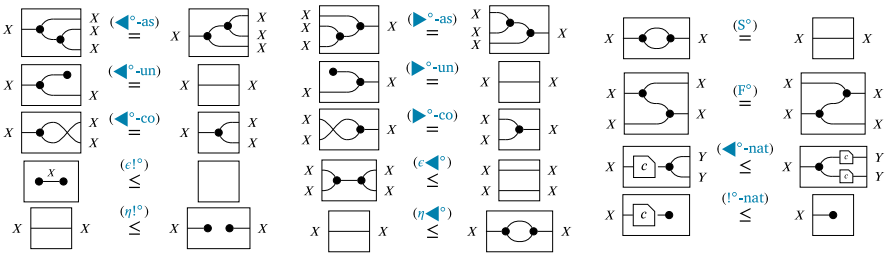


Fig. 2. Axioms of cartesian bicategories.

2. every arrow $c : X \to Y$ is a *lax comonoid homomorphism*, i.e., the inequalities ($\blacktriangleleft^\circ$-nat) and (!°-nat) in Fig. 2 hold;
3. comonoids are *left adjoints* to the monoids, i.e., the inequalities ($\eta\blacktriangleleft^\circ$), ($\epsilon\blacktriangleleft^\circ$), ($\eta!^\circ$) and ($\epsilon!^\circ$) in Fig. 2 hold;
4. monoids and comonoids form *special Frobenius bimonoids*, i.e., the equalities (F°) and (S°) in Fig. 2 hold;
5. monoids and comonoids satisfy the expected coherence conditions (see e.g. [46]).

A *morphism of cartesian bicategories* is a poset-enriched strong symmetric monoidal functor preserving monoids and comonoids. We denote by $\mathbb{CB}$ the category of cartesian bicategories and their morphisms.

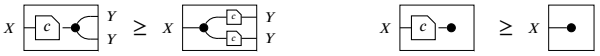
As already mentioned, **Rel**° with $\blacktriangleleft^\circ_X$, $!^\circ_X$, $\blacktriangleright^\circ_X$ and $¡^\circ_X$ defined in (3) form a cartesian bicategory: the reader can easily check, using the definitions in (1) and (3), that all the laws in Fig. 2 are satisfied. For instance, checking the property of lax comonoid homomorphism in **Rel**° amounts to observe that the following inclusions hold for all relations $c \subseteq X \times Y$.

$$\{(x,(y,y)) \mid (x,y) \in c\} \subseteq \{(x,(y_1,y_2)) \mid (x,y_1) \in c \wedge (x,y_2) \in c\} \qquad (\blacktriangleleft^\circ\text{-nat})$$

$$\{(x,\star) \mid \exists y \in Y.\,(x,y) \in c\} \subseteq \{(x,\star) \mid x \in X\} \qquad (!^\circ\text{-nat})$$

Interestingly, one can observe that the opposite inequality of ($\blacktriangleleft^\circ$-nat) holds iff the relation $c \subseteq X \times Y$ is single-valued (i.e., deterministic), while the opposite of (!°-nat) iff c is total. In other words, the relation c is a function iff both ($\blacktriangleleft^\circ$-nat) and (!°-nat) hold as equalities. This observation leads to generalise the notion of functions from **Rel**° to the notion of map in an arbitrary cartesian bicategory.

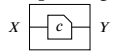
Definition 2.2. Let $c : X \to Y$ be an arrow of a cartesian bicategory. We say that c is *single valued* if the leftmost inequality below holds; c is *total* if the rightmost inequality holds; c is a *map* if it is both single valued and total.



(4)

Moreover, we say that c is *injective* if the leftmost inequality below holds, and c is *surjective* if the rightmost inequality holds.



In several occasions, the properties of maps will play a fundamental role. Therefore we adopt the graphical convention to draw maps $c : X \to Y$ as  to distinguish them for arbitrary arrows that are drawn as  . It is easy to see that maps form a monoidal subcategory of **C** [44], hereafter denoted by $\mathsf{Map}(\mathbf{C})$. Since, by ($\blacktriangleleft^\circ$-nat), (!°-nat) and (4), comonoids are natural w.r.t. maps, i.e.,

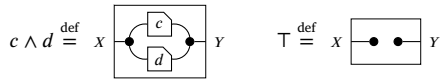


Fox's theorem [50] guarantees that $\mathsf{Map}(\mathbf{C})$ is a *category with finite products*.

Proposition 2.1. *For any cartesian bicategory* **C**, $\mathsf{Map}(\mathbf{C})$ *is a category with finite products. Moreover, a morphism of cartesian bicategories* $F : \mathbf{C} \to \mathbf{D}$ *restricts to a functor* $\bar{F} : \mathsf{Map}(\mathbf{C}) \to \mathsf{Map}(\mathbf{D})$ *preserving finite products.*

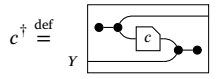
Proof. See e.g. Proposition 2.44 in [55]. □

In a cartesian bicategory **C**, each homset $\mathbf{C}[X,Y]$ carries the structure of inf-semilattice, defined for all $c,d : X \to Y$ as follows.



(5)

As expected in **Rel**°, $c \wedge d$ is the intersection of the relations c and d and $\top$ is the top relation $\{(x,y) \mid x \in X,\ y \in Y\}$ (the reader can check using (1), (3)). Furthermore, the Eq. (6) below defines an identity-on-objects isomorphism of cartesian bicategories $(\cdot)^\dagger : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$, satisfying the properties in Table 1.



(hereafter drawn as)

(6)

Table 1
Properties of $(\cdot)^\dagger : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$.

if $c \leq d$ then $c^\dagger \leq d^\dagger$		$(c^\dagger)^\dagger = c$	
$(c \fatsemi d)^\dagger = d^\dagger \fatsemi c^\dagger$	$(id^\circ_X)^\dagger = id^\circ_X$	$(\blacktriangleright^\circ_X)^\dagger = \blacktriangleleft^\circ_X$	$(\mathrm{i}^\circ_X)^\dagger = !^\circ_X$
$(c \otimes d)^\dagger = c^\dagger \otimes d^\dagger$	$(\sigma^\circ_{X,Y})^\dagger = \sigma^\circ_{Y,X}$	$(\blacktriangleleft^\circ_X)^\dagger = \blacktriangleright^\circ_X$	$(!^\circ_X)^\dagger = \mathrm{i}^\circ_X$
$(c \wedge d)^\dagger = c^\dagger \wedge d^\dagger$		$\top^\dagger = \top$	

The reader can check, using (1) and (3) that in $\mathbf{Rel}^\circ$, $c^\dagger : Y \to X$ is the opposite of the relation c, namely $\{(y,x) \mid (x,y) \in c\}$. It is well known that a relation c is a function iff it is left adjoint to $c^\dagger$. More generally in a cartesian bicategory c is a map iff it is left adjoint to $c^\dagger$.

The discussion above is summarised by the following proposition.

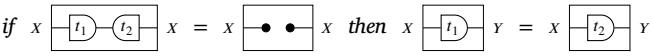
Proposition 2.2. *Let* $\mathbf{C}$ *be a cartesian bicategory and* $c : X \to Y$ *an arrow of* $\mathbf{C}$*. The following hold:*

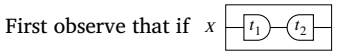
1. *every homset carries the inf-semilattice structure, defined as in* (5)*;*
2. *there is an isomorphism of cartesian bicategories* $(\cdot)^\dagger : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$*, defined as in* (6)*, enjoying the properties in Table 1.*
3. *c is a map iff c is left adjoint to $c^\dagger$, namely $id^\circ_X \leq c \fatsemi c^\dagger$ and $c^\dagger \fatsemi c \leq id^\circ_Y$;*

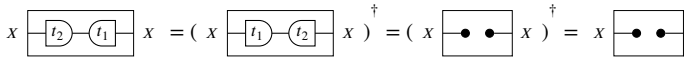
Proof. For 1. see e.g. Corollary 2.37 in [55]. For 2. see Theorem 2.4 in [44]. For 3. see Lemma 2.5 in [44]. □

The following result marks a key difference between cartesian bicategories and hyperdoctrines that we will discuss later in Section 4.4.

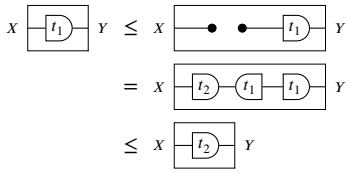
Proposition 2.3 (Extensional equality). *For any* $t_1, t_2 : X \to Y$ *maps,*

if  *then* 

Proof. First observe that if  , then by the properties in Table 1 it holds that

 (7)

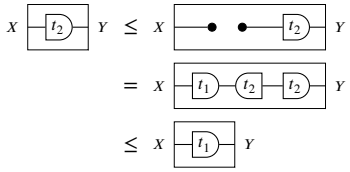
To conclude we show the two inclusions separately. For the left-to-right inclusion the following holds:


($\eta!^\circ$)

$=$ (7)

$\leq$ (Proposition 2.2.3)

The other inclusion is proved as follows:


($\eta!^\circ$)

$=$ (Hypothesis)

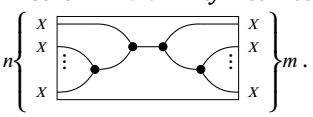
$\leq$ (Proposition 2.2.3)

□

Remark 2.1. Note that, similar to Peirce's calculus of relations, reasoning with string diagrams proceeds (in)equationally: one substitutes a subdiagram with another that contains it or is equal to it, as illustrated, for example, in the proof above.

Before introducing cocartesian bicategories, we conclude this subsection by listing several properties of cartesian bicategories that will be useful later in the technical development of the paper.

Theorem 2.1. *Any* connected *diagram* $c : X^n \to X^m$ *made out of* $id^\circ, \sigma^\circ, \blacktriangleleft^\circ, !^\circ, \blacktriangleright^\circ$ *and* i° *is equivalent to*



Proof. See [56,57]. □

Remark 2.2. Theorem 2.1 is known as the *spider theorem* and it holds whenever objects carry a special Frobenius algebra. A direct consequence of the spider theorem is that we can "rewire" parts of a diagram involving the (co)monoid structures, as long as the connectivity is preserved. This is useful in several graphical derivations for arranging diagrams into a desired shape.

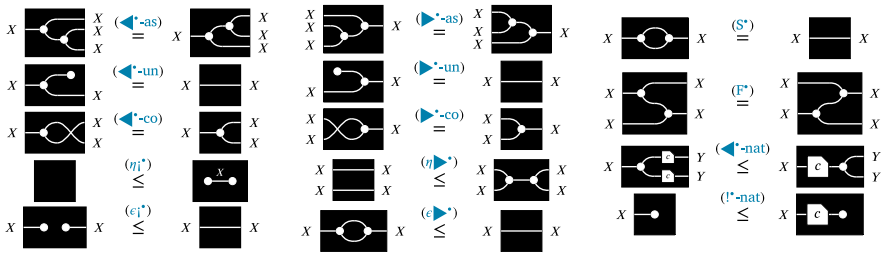
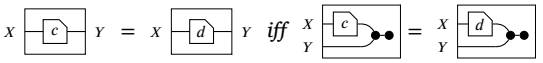
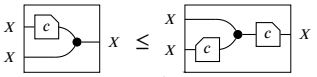
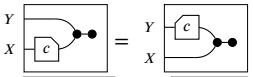
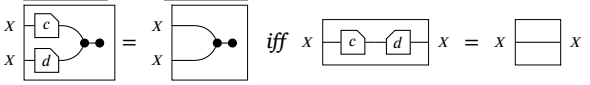


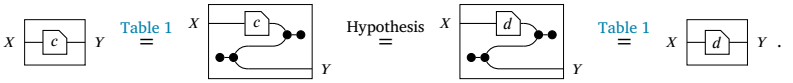
Fig. 3. Axioms of cocartesian bicategories.

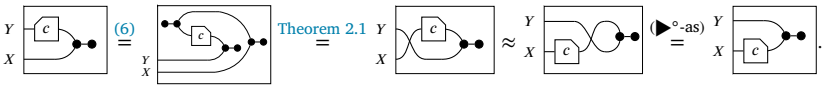
Lemma 2.1. *For any $c, d : X \to Y$, the following hold:*

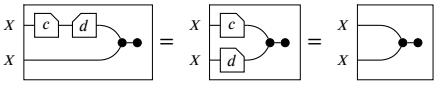
1. 
2. 
3. 
4. 

Proof. We prove each statement below.

1. The left-to-right direction is trivial. The other direction is proved as follows:



2. See Lemma 4.3 in [58].
3. 
4. By 3. it holds that



and thus we conclude by 1. □

In the derivation above, and in many of the graphical derivations illustrated hereafter, we use ≈ for indicating that the equivalence holds by the axioms of symmetric monoidal categories.

2.2. Cocartesian bicategories

Cocartesian bicategories are just cartesian bicategories where the preorder is reversed. For the convenience of the reader, we reported their diagrammatic axioms in Fig. 3, but their definition simply amounts to the following.

Definition 2.3. $\mathbf{C}$ is a *cocartesian bicategory* if $\mathbf{C}^{\mathrm{co}}$ is a cartesian bicategory. A *morphism of (co)cartesian bicategories* is a poset-enriched strong symmetric monoidal functor preserving monoids and comonoids.

Our main example is $\mathbf{Rel}^\bullet$ with monoids and comonoids defined as $\blacktriangleleft^\bullet$, $\blacktriangleright^\bullet$, $!^\bullet$ and $¡^\bullet$ in (3). To see that $\mathbf{Rel}^\bullet$ is a cocartesian bicategory, observe that the complement $\neg(\cdot)$ is a poset-enriched symmetric monoidal isomorphism $\neg(\cdot) : (\mathbf{Rel}^{\circ})^{\mathrm{co}} \to \mathbf{Rel}^\bullet$ preserving (co)monoids.

It is not surprising that in a cocartesian bicategory $\mathbf{C}$, every homset $\mathbf{C}[X, Y]$ carries a sup semi-lattice, where $c \vee d$ and $\bot$ are defined for all arrows $c, d : X \to Y$ as follows.



(8)

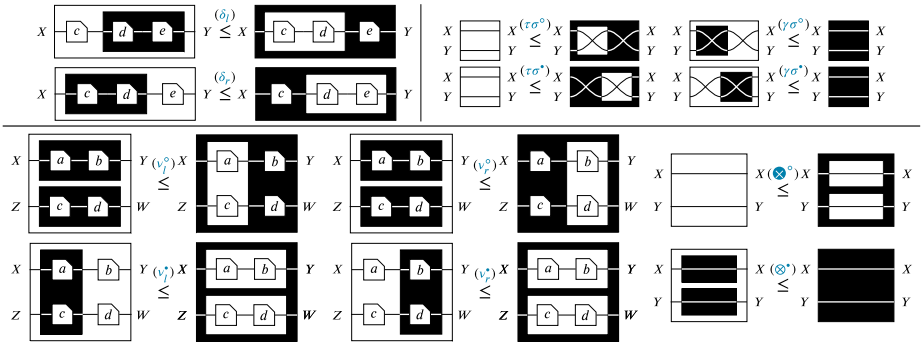


Fig. 4. Axioms of closed symmetric monoidal linear bicategories.

Similarly, for any cocartesian bicategory $\mathbf{C}$, there is an identity on objects isomorphism $(\cdot)^{\ddagger} : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$ defined as follows:



$$c^{\ddagger} \stackrel{\text{def}}{=} \tag{9}$$

2.3. Linear bicategories

We have seen that $\mathbf{Rel}^{\circ}$ forms a cartesian bicategory, and $\mathbf{Rel}^{\bullet}$ a cocartesian bicategory. The next step consists in merging them into one entity and study their algebraic interactions. However, the coexistence of two different compositions $\fatsemi$ and $;$ on the same class of objects and arrows brings us out of the realm of ordinary categories. The appropriate setting is provided by *linear bicategories* [45] by Cockett, Koslowski and Seely.

Definition 2.4. A *linear bicategory* $(\mathbf{C}, \fatsemi, id^{\circ}, ;, id^{\bullet})$ consists of two poset-enriched categories $(\mathbf{C}, \fatsemi, id^{\circ})$ and $(\mathbf{C}, ;, id^{\bullet})$ with the same class of objects, arrows and orderings (but possibly different identities and compositions) such that

1. $\fatsemi$ linearly distributes over $;$, i.e., (δ_l) and (δ_r) in Fig. 4 hold.

A *symmetric monoidal linear bicategory* $(\mathbf{C}, \fatsemi, id^{\circ}, ;, id^{\bullet}, \otimes, \sigma^{\circ}, \boxtimes, \sigma^{\bullet}, I)$, shortly $(\mathbf{C}, \otimes, \boxtimes, I)$, consists of a linear bicategory $(\mathbf{C}, \fatsemi, id^{\bullet}, ;, id^{\bullet})$ and two poset-enriched symmetric monoidal categories $(\mathbf{C}, \otimes, I)$ and $(\mathbf{C}, \boxtimes, I)$ s.t. $\otimes$ and $\boxtimes$ agree on objects, i.e., $X \otimes Y = X \boxtimes Y$, share the same unit I and

2. there are linear strengths for $(\otimes, \boxtimes)$, i.e., the inequalities (ν_l°), (ν_r°), $(\nu_l^{\bullet})$ and $(\nu_r^{\bullet})$ in Fig. 4 hold;
3. $\boxtimes$ preserves id° colaxly and $\otimes$ preserves $id^{\bullet}$ laxly, i.e., $(\otimes^{\bullet})$ and $(\boxtimes^{\circ})$ hold.

A *morphism of symmetric monoidal linear bicategories* $F : (\mathbf{C_1}, \otimes, \boxtimes, I) \to (\mathbf{C_2}, \otimes, \boxtimes, I)$ consists of two poset-enriched symmetric monoidal functors $F^{\circ} : (\mathbf{C_1}, \otimes, I) \to (\mathbf{C_2}, \otimes, I)$ and $F^{\bullet} : (\mathbf{C_1}, \boxtimes, I) \to (\mathbf{C_2}, \boxtimes, I)$ that agree on objects and arrows: $F^{\circ}(X) = F^{\bullet}(X)$ and $F^{\circ}(c) = F^{\bullet}(c)$.

All linear bicategories in this paper are symmetric monoidal. Hence, we usually omit the adjective *symmetric monoidal* and refer to them simply as linear bicategories. For a linear bicategory $(\mathbf{C}, \otimes, \boxtimes, I)$, we will often refer to $(\mathbf{C}, \otimes, I)$ as the *white structure*, shorthand $\mathbf{C}^{\circ}$, and to $(\mathbf{C}, \boxtimes, I)$ as the *black structure*, $\mathbf{C}^{\bullet}$. Note that a morphism F is a mapping of objects and arrows that preserves the ordering, the white and black structures; thus we write F for both F° and $F^{\bullet}$.

Definition 2.5. Let $a : X \to Y$ and $b : Y \to X$ be arrows of a linear bicategory. The arrow a is *left linear adjoint* to b, in symbols $b \Vdash a$, iff

$$id^{\circ}_X \leq a ; b \text{ and } b \fatsemi a \leq id^{\bullet}_Y .$$

Definition 2.6. A linear bicategory $(\mathbf{C}, \otimes, \boxtimes, I)$ is said to be *closed* if every $a : X \to Y$ has both a left and a right linear adjoint and, in particular, the white symmetry σ° is both left and right linear adjoint to the black symmetry $\sigma^{\bullet}$ ($\sigma^{\bullet} \Vdash \sigma^{\circ} \Vdash \sigma^{\bullet}$), i.e. $(\tau\sigma^{\circ})$, $(\gamma\sigma^{\circ})$, $(\tau\sigma^{\bullet})$ and $(\gamma\sigma^{\bullet})$ in Fig. 4 hold.

Our main example is the closed linear bicategory **Rel** of sets and relations. The white structure is the symmetric monoidal category $\mathbf{Rel}^{\circ}$ and the black structure is $\mathbf{Rel}^{\bullet}$. Observe that the two have the same objects, arrows and ordering. The white and black monoidal products $\otimes$ and $\boxtimes$ agree on objects (they are the cartesian product of sets) and have common unit object (the singleton set I). By (1) and (2), one can easily check all the inequalities in Fig. 4. Both left and right linear adjoints of a relation c are given by $\neg c^{\dagger}$.

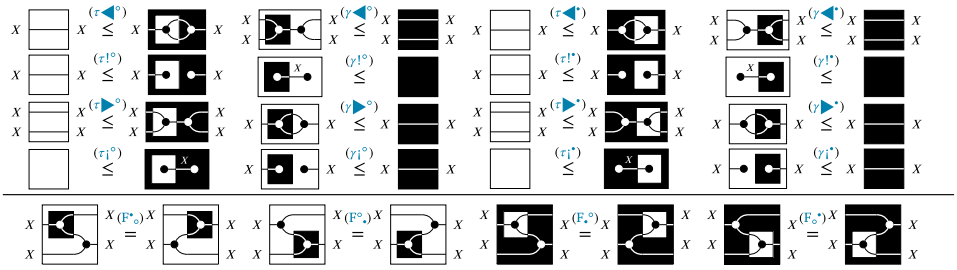


Fig. 5. Additional axioms for fo-bicategories.

2.4. First-order bicategories

After cartesian, cocartesian and linear bicategories, we can now recall first-order bicategories from [43].

Definition 2.7. A *first-order bicategory* $\mathbf{C}$ consists of a closed linear bicategory $(\mathbf{C}, \otimes, \oplus, I)$, a cartesian bicategory $(\mathbf{C}, \blacktriangleleft^{\circ}, \blacktriangleright^{\circ})$ and a cocartesian bicategory $(\mathbf{C}, \blacktriangleleft^{\bullet}, \blacktriangleright^{\bullet})$, such that

1. the white comonoid $(\blacktriangleleft^{\circ}, !^{\circ})$ is left and right linear adjoint to black monoid $(\blacktriangleright^{\bullet}, ¡^{\bullet})$ and the white monoid $(\blacktriangleright^{\circ}, ¡^{\circ})$ is left and right linear adjoint to black comonoid $(\blacktriangleleft^{\bullet}, !^{\bullet})$ i.e., the 16 inequalities in the top of Fig. 5 hold;
2. white and black (co)monoids satisfy the linear Frobenius laws, i.e. the equalities $(F^{\bullet}_{\circ})$, $(F^{\circ}_{\bullet})$, $(F^{\circ}_{\circ})$, $(F^{\bullet}_{\bullet})$ in Fig. 5 hold.

A *morphism of fo-bicategories* is a morphism of linear bicategories *and* of (co)cartesian bicategories. We denote by $\mathbb{FOB}$ the category of fo-bicategories and their morphisms.

We have seen that **Rel** is a closed linear bicategory, $\mathbf{Rel}^{\circ}$ a cartesian bicategory and $\mathbf{Rel}^{\bullet}$ a cocartesian bicategory. Given (3), it is easy to check the inequalities in Fig. 5.

Whenever $\mathbf{C}$ is a fo-bicategory, then $\mathbf{C}^{\mathrm{co}}$ is a fo-bicategory when swapping white and black structures. Similarly, $\mathbf{C}^{\mathrm{op}}$ is a fo-bicategory when swapping monoids and comonoids. In any fo-bicategory, left and right linear adjoints of an arrow c coincide and are denoted by $c^{\perp}$. The assignment $c \mapsto c^{\perp}$ gives rise to an identity-on-objects isomorphism of fo-bicategories $(\cdot)^{\perp} : \mathbf{C} \to (\mathbf{C}^{\mathrm{co}})^{\mathrm{op}}$. Similarly, $(\cdot)^{\dagger} : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$ in (6) is also an isomorphism of fo-bicategories. Since the diagram below commutes,

$$\begin{array}{ccc} \mathbf{C} & \xrightarrow{(\cdot)^{\dagger}} & \mathbf{C}^{\mathrm{op}} \\ {\scriptstyle (\cdot)^{\perp}}\downarrow & & \downarrow{\scriptstyle (\cdot)^{\perp}} \\ (\mathbf{C}^{\mathrm{co}})^{\mathrm{op}} & \xrightarrow{(\cdot)^{\dagger}} & \mathbf{C}^{\mathrm{co}} \end{array}$$

one can define the complement as the diagonal of the square, namely

$$\neg(\cdot) \stackrel{\mathrm{def}}{=} ((\cdot)^{\perp})^{\dagger}.$$

Clearly $\neg(\cdot) : \mathbf{C} \to \mathbf{C}^{\mathrm{co}}$ is an isomorphism of fo-bicategories. Moreover, it induces a boolean algebra on each homset of $\mathbf{C}$.

Proposition 2.4. *Let* $\mathbf{C}$ *be a fo-bicategory. Then, every homset of* $\mathbf{C}$ *is a boolean algebra.*

Proof. See Proposition 4.23 in [55]. □

Proposition 2.5. *Let* $F : \mathbf{C} \to \mathbf{D}$ *be a morphism of fo-bicategories. Then,* $\neg F(c) = F(\neg c)$ *for all arrows* c*, and hence* F *preserves the boolean structure on the homsets.*

Proof. Follows from the definition of $\neg(\cdot)$ and the fact that $F(c)^{\dagger} = F(c^{\dagger})$ (see Lemma 2.35 in [55]) and $F(c)^{\perp} = F(c^{\perp})$ (see Lemma 4.16 in [55]). □

The next property of maps (Definition 2.2) plays a key role in our work.

Proposition 2.6. *For all maps* $f : X \to Y$ *and arrows* $c : Y \to Z$*, it holds that*

$$f \fatsemi \neg c = \neg(f \fatsemi c).$$

Proof. Follows from Proposition 4.22 in [55], the definition of $\neg(\cdot)$ and the fact that $(\cdot)^{\dagger}$ and $(\cdot)^{\perp}$ are isomorphisms of fo-bicategories. □

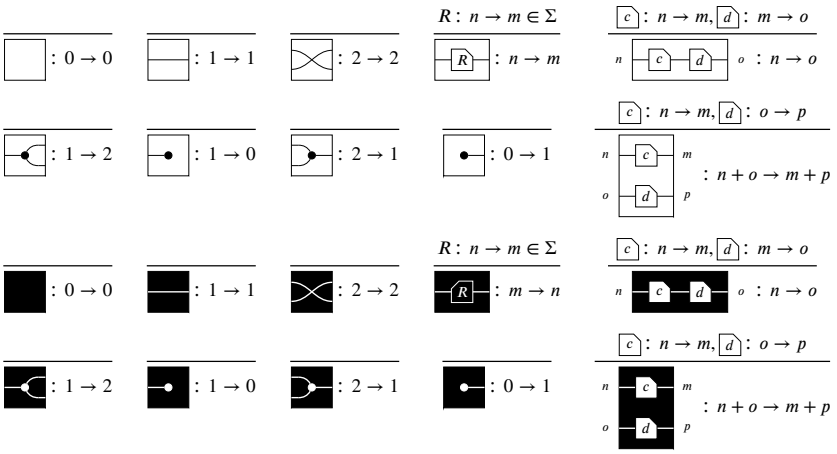
In the fo-bicategory **Rel**, the above equalities amounts to observe that the following derivation holds for all functions $f : X \to Y$ and relations $c \subseteq Y \times Z$.

$$\begin{aligned} \{(x,z) \mid \exists y.\, f(x) = y \wedge (y,z) \in \neg c\} &= \{(x,z) \mid (f(x),z) \in \neg c\} \\ &= \neg\{(x,z) \mid (f(x),z) \in c\} \\ &= \neg\{(x,z) \mid \exists y.\, f(x) = y \wedge (y,z) \in c\} \end{aligned}$$

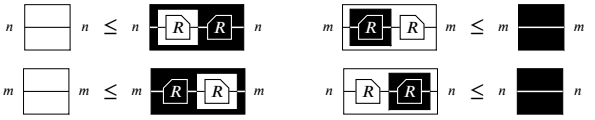
2.5. Freely generated first-order bicategories

We conclude this section by giving to the reader a taste of how fo-bicategories relate to first-order theories. First, we recall from [43] the freely generated fo-bicategory $\mathbf{FOB}_\Sigma$.

Given a monoidal signature Σ, namely a set of symbols $R : n \to m$ with arity n and coarity m, $\mathbf{FOB}_\Sigma$ is the fo-bicategory whose objects are natural numbers and arrows $c : n \to m$ are string diagrams generated by the following rules:



More precisely, arrows are equivalence classes of string diagrams w.r.t $\lesssim \cap \gtrsim$, where $\lesssim$ is the precongruence (w.r.t. $\fatsemi, \otimes, +$ and $\otimes$) generated by the axioms in Figs. 2–5 (with X, Y, Z, W replaced by natural numbers, and a, b, c, d by diagrams of the appropriate type) and the axioms forcing  and  to be linear adjoints:

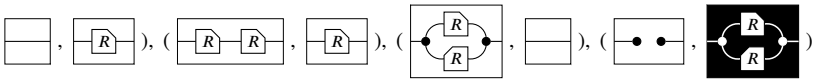


To give semantics to these diagrams we need *interpretations*, i.e. pairs $\mathcal{I} = (X, \rho)$, where X is a set and ρ is a function assigning to each $R : n \to m \in \Sigma$ a relation $\rho(R) : X^n \to X^m$. Since $\mathbf{FOB}_\Sigma$ is the free fo-bicategory, for any interpretation $\mathcal{I}$ there exists a unique morphism of fo-bicategories $\mathcal{I}^\sharp : \mathbf{FOB}_\Sigma \to \mathbf{Rel}$ such that $\mathcal{I}^\sharp(1) = X$ and $\mathcal{I}^\sharp($  $) = \rho(R) \subseteq X^n \times X^m$. Intuitively, $\mathcal{I}^\sharp$ is defined inductively by (1), (2) and (3) with the free cases provided by $\mathcal{I}$.

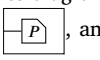
Definition 2.8. A *diagrammatic first-order theory* is a pair $\mathbb{T} = (\Sigma, \mathbb{I})$ where Σ is a monoidal signature and $\mathbb{I}$ is a set of *axioms*: pairs (c, d) for $c, d : n \to m$ in $\mathbf{FOB}_\Sigma$, standing for $c \leq d$. An interpretation $\mathcal{I}$ is a *model* of $\mathbb{T}$ if and only if, for all $(c, d) \in \mathbb{I}$, $\mathcal{I}^\sharp(c) \subseteq \mathcal{I}^\sharp(d)$.

As illustrated in [43], one can generate the fo-bicategory $\mathbf{FOB}_\mathbb{T}$ and, in the spirit of Lawvere's functorial semantics [59], models of $\mathbb{T}$ are in one-to-one correspondence with morphisms $F : \mathbf{FOB}_\mathbb{T} \to \mathbf{Rel}$.

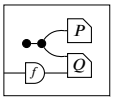
Example 2.1. Consider the theory $\mathbb{T} = (\Sigma, \mathbb{I})$, where $\Sigma = \{R : 1 \to 1\}$ and $\mathbb{I}$ be as follows:



An interpretation is a set X and a relation $R \subseteq X \times X$. It is a model iff R is an order, i.e., reflexive ($id^\circ_X \subseteq R$), transitive ($R \fatsemi R \subseteq R$), antisymmetric ($R \cap R^\dagger \subseteq id^\circ$) and total ($\top \subseteq R \cup R^\dagger$).

Remark 2.3. A direct encoding of traditional first-order theories into diagrammatic ones is illustrated in [43]. Shortly, a predicate symbol P of arity n becomes a symbol $P : n \to 0 \in \Sigma$, drawn as  , and a n-ary function symbol f becomes $f : n \to 1 \in \Sigma$, drawn as  .

For instance, the formula $\exists x.P(x) \wedge Q(x, f(y))$ is rendered below, where  plays the role of $\exists$ and  that of $\wedge$.



Note that both predicate and function symbols of traditional first-order theories are regarded as symbols of the monoidal signature Σ. Function symbols are constrained to represent functions by adding to $\mathbb{I}$ the axioms of maps, i.e., the inequalities in (4).

3. Peircean bicategories

In this section we introduce the notion of *peircean bicategory*, and we prove that such a new notion provides an alternative presentation of fo-bicategories. The name peircean is due to the fact that, like in Peirce's algebra of relations [54], and differently from fo-bicategories, the structure of boolean algebra is taken as a primitive.

Definition 3.1. A *peircean bicategory* consists of a cartesian bicategory $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ such that

1. every homset $\mathbf{C}[X, Y]$ carries a boolean algebra $(\mathbf{C}[X, Y], \vee, \bot, \wedge, \top, \neg)$;
2. for all maps $f : X \to Y$ and arrows $c : Y \to Z$,

$$f \fatsemi \neg c = \neg(f \fatsemi c). \tag{$\neg\mathcal{M}$}$$

A *morphism of peircean bicategories* is a morphism of cartesian bicategories $F : \mathbf{C} \to \mathbf{D}$ such that $F(\neg c) = \neg F(c)$. We write $\mathbb{PB}$ for the category of peircean bicategories and their morphisms.

Note that, differently from Definition 2.7, Definition 3.1 is not purely axiomatic, since the property 2 requires f to be a map. However, the notion of a peircean bicategory is notably more succinct than that of a fo-bicategory, making it more convenient for illustrating the correspondence with boolean hyperdoctrines in Section 5.2.

By Propositions 2.4 and 2.6, every fo-bicategory is a peircean bicategory. By Proposition 2.5 every morphism of fo-bicategories is a morphism of peircean bicategories.

Vice versa, every peircean bicategory $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ gives rise to a fo-bicategory. The black structure $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$ is defined as expected from the white one and $\neg$:

$$\begin{array}{llll} c \fatsemi d \overset{\text{def}}{=} \neg(\neg c \fatsemi \neg d) & id^\bullet_X \overset{\text{def}}{=} \neg id^\circ_X & c \otimes d \overset{\text{def}}{=} \neg(\neg c \otimes \neg d) & \sigma^\bullet_{X,Y} \overset{\text{def}}{=} \neg\sigma^\circ_{X,Y} \\ \blacktriangleleft^\bullet_X \overset{\text{def}}{=} \neg\blacktriangleleft^\circ_X & !^\bullet_X \overset{\text{def}}{=} \neg!^\circ_X & \blacktriangleright^\bullet_X \overset{\text{def}}{=} \neg\blacktriangleright^\circ_X & \text{¡}^\bullet_X \overset{\text{def}}{=} \neg\text{¡}^\circ_X \end{array} \tag{10}$$

With this definition, it is immediate to see that $\neg : (\mathbf{C}^{\mathrm{co}}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet) \to (\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ is an isomorphism of cartesian bicategories and thus to conclude that $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$ is a cocartesian bicategory.

Proving that $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ and $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$ give rise to a fo-bicategory is the main technical effort of this paper. The remainder of this section is dedicated to proving the following result.

Theorem 3.1. *There is an isomorphism of categories* $\mathbb{FOB} \cong \mathbb{PB}$.

3.1. Proof of Theorem 3.1

The core of the proof involves demonstrating that the axioms in Figs. 4 and 5 hold with the structure defined in (10). Before delving into the main proofs, we need to establish some auxiliary results, whose proofs are reported in A. There, and in most of the following graphical derivations, for a generic arrow $\boxed{c}$, we depict its negation as $\blacksquare c$ and, using the graphical convention in (6), $(\neg\boxed{c})^\dagger$ is depicted as $\blacksquare c$.

To prove inclusions between arrows, we will find often convenient to use one of the following characterisations provided by the properties of boolean algebras.

Proposition 3.1. *For any $c, d : X \to Y$ the following are equivalent:*

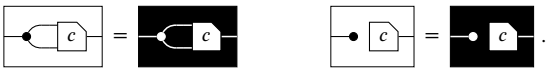
$$1.\ c \leq d \qquad 2.\ \neg d \leq \neg c \qquad 3.\ \top \leq \neg c \vee d \qquad 4.\ c \wedge \neg d \leq \bot$$

We will also find convenient to use the following reformulation of $(\neg\mathcal{M})$.

Proposition 3.2. *For all maps $f : X \to Y$ and arrows $c : Y \to Z$ it holds that*

$$f \fatsemi c = \neg f \fatsemi c. \tag{$\bullet\mathcal{M}$}$$

When taking as f the maps $\blacktriangleleft^\circ$ or $!^\circ$ we have the following equalities.



By definition in a peircean bicategory $\mathbf{C}$, each homset $\mathbf{C}[X, Y]$ carries a boolean algebra. The inf-semilattice structure coincides, by definition, with the one of the cartesian bicategory $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ as defined in (5). By (10), the sup-semilattice structure coincide with the one of the cocartesian bicategory $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$ as defined in (8).

Lemma 3.1. *For all arrows $c, d : X \to Y$, it holds that*

$$c \vee d = \blacktriangleleft^\bullet_X \fatsemi (c \otimes d) \fatsemi \blacktriangleright^\bullet_Y \quad \text{and} \quad \bot = !^\bullet_X \fatsemi \text{¡}^\bullet_Y,$$

graphically rendered as in (8).

Since $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ is a cartesian bicategory and $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$ a cocartesian one, there are two isomorphisms $(\cdot)^\dagger, (\cdot)^\ddagger : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$ defined as in (6) and in (9), respectively. By means of (10), one can prove the following equalities.

Lemma 3.2. *For any* $c : X \to Y$*, it holds that* $(\neg c)^\dagger = \neg(c^\dagger)$.

Lemma 3.3. *For any* $c : X \to Y$*, it holds that* $c^\dagger = c^\ddagger$.

Lemma 3.3 is instrumental in proving the following result that extends $(\neg\mathcal{M})$ and therefore $(\bullet\mathcal{M})$, to *comaps*, namely all those arrows f, such that $f^\dagger$ is a map.

Lemma 3.4. *For all maps* $f : X \to Y$ *and arrows* $c : Z \to Y$*, it holds that*

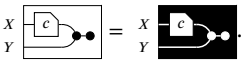
$$\neg c \fatsemi f^\dagger = \neg(c \fatsemi f^\dagger) \quad \text{and} \quad c \fatsemi f^\dagger = c \fatsemi \neg f^\dagger.$$

It is convenient to visualize in diagrams Lemma 3.4 on two particular cases, namely when we take as comap $\blacktriangleright^\circ$ or ¡°:



Lastly, observe that the structure in (10) can also be used to define a compact closed structure in $(\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$. In other words, in a peircean bicategory one can *bend* the wires in two ways. These bending operations are shown to be the same.

Lemma 3.5. *For any* $c : X \to Y$*, it holds that*



Having established these preliminary results, we now need to prove that the axioms of fo-bicategories hold in a peircean bicategory. We do not show a proof for all of them, but only for a few representative ones. The rest are either derivable or proved in a similar manner.

Linear distributivities. First, we prove that linear distributivities hold for $\otimes$ and $\oslash$, leveraging on the same property for $\wedge$ over $\vee$. Then, we show a form of linear distributivity for arrows of type $X \to I$, enabling a streamlined proof for the general statement.

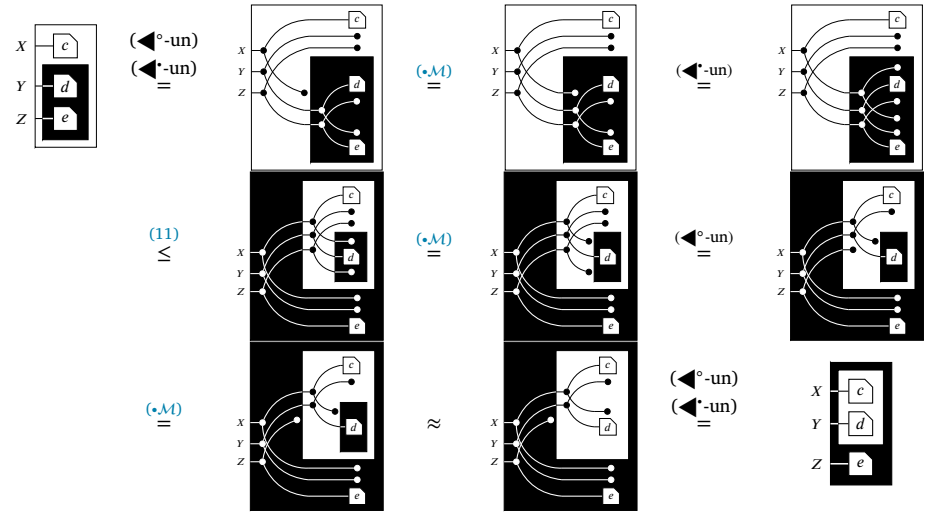
Lemma 3.6. *For any* $c : X \to I, d : Y \to I, e : Z \to I$*, it holds that*

$$c \otimes (d \oslash e) \leq (c \otimes d) \oslash e.$$

Proof. Since $\mathbf{C}[X, Y]$ is a boolean algebra, for any $c, d, e : X \to Y$ it holds that

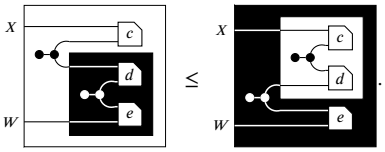
$$c \wedge (d \vee e) \leq (c \wedge d) \vee e. \tag{11}$$

To conclude, it suffices to put the statement in the form of (11) as follows.

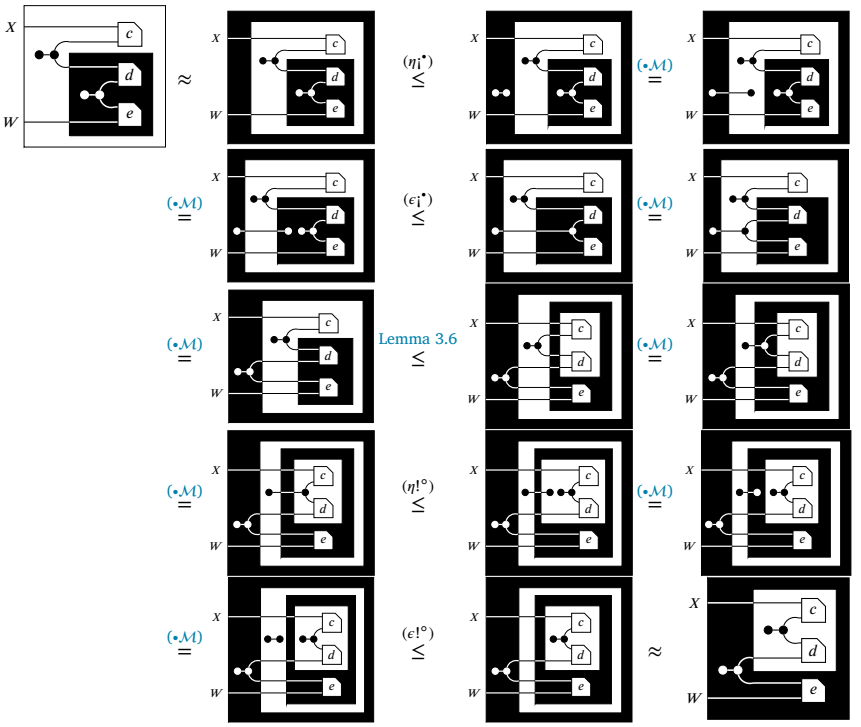


□

Lemma 3.7. *For any* $c : X \otimes Y \to I, d : Y \otimes Z \to I, e : Z \otimes W \to I$*, it holds that*



Proof.

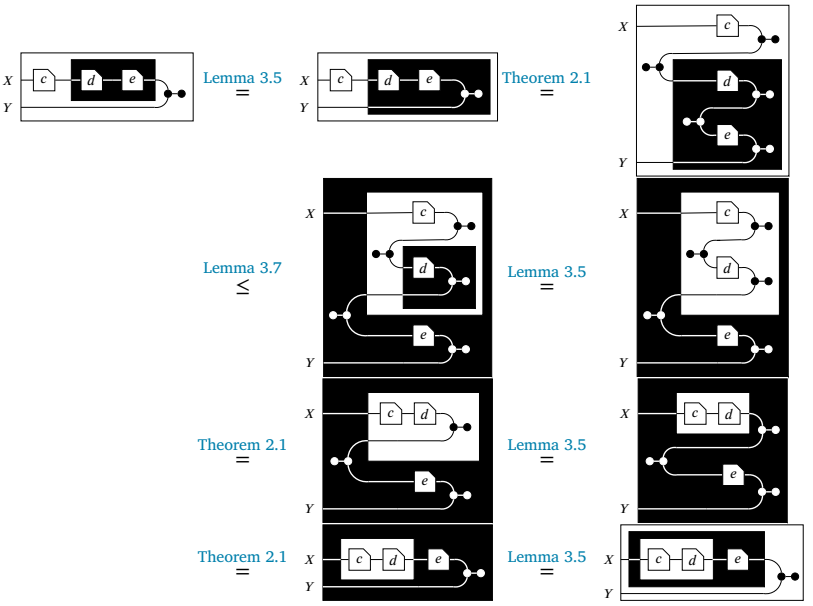


□

Lemma 3.8. *For any* $c : X \to Y, d : Y \to Z, e : Z \to W$*, it holds that*

$$c \fatsemi (d ; e) \leq (c \fatsemi d) ; e \quad \text{and} \quad (c ; d) \fatsemi e \leq c ; (d \fatsemi e).$$

Proof. Observe that the following holds:



and thus by Lemma 2.1.1 we have that $c \fatsemi (d ; e) \leq (c \fatsemi d) ; e$. The other inequality is proved analogously. □

Linear adjoints. We now prove that in a peircean bicategory $\neg c^\dagger \dashv\vdash c$ and $c \dashv\vdash \neg c^\dagger$, namely that $\neg c^\dagger$ is both left and right linear adjoint to c.

Lemma 3.9. *For any $c : X \to Y$, it holds that $\neg c^\dagger$ is left linear adjoint to c, namely*

$$c \fatsemi (\neg c)^\dagger \leq id^\bullet_X \quad \text{and} \quad id^\circ_Y \leq (\neg c)^\dagger \fatsemi c,$$

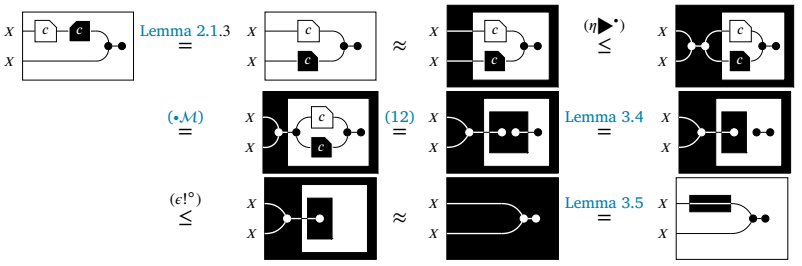
and $\neg c^\dagger$ is right linear adjoint to c, namely

$$(\neg c)^\dagger \fatsemi c \leq id^\bullet_Y \quad \text{and} \quad id^\circ_X \leq c \fatsemi (\neg c)^\dagger.$$

Proof. Since $\mathbf{C}[X, Y]$ is a boolean algebra, for any arrow $c : X \to Y$ we have that

$$c \wedge \neg c = \bot \quad \text{and} \quad c \vee \neg c = \top. \tag{12}$$

Now observe that the following holds:



and thus by Lemma 2.1.1 we have that $c \fatsemi (\neg c)^\dagger \leq id^\bullet_X$. The other inequality, i.e. $id^\circ_Y \leq (\neg c)^\dagger \fatsemi c$, is proved analogously, using the other equality in (12).

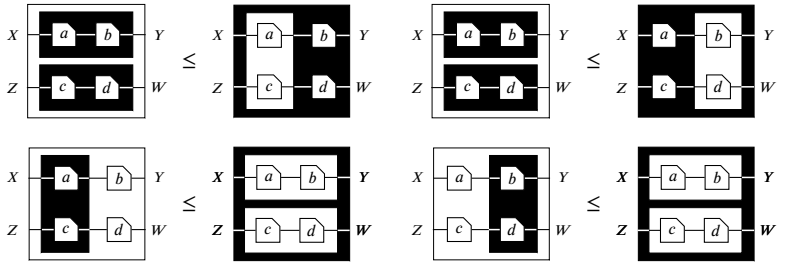
This proves that $\neg c^\dagger$ is left linear adjoint to c. To prove that it is also the right linear adjoint we use the properties of $\neg(\cdot)$ and $(\cdot)^\dagger$:

$$\begin{aligned}
id^\circ_Y \leq (\neg c)^\dagger \fatsemi c &\iff \neg((\neg c)^\dagger \fatsemi c) \leq \neg id^\circ_Y && \text{(Proposition 3.1)}\\
&\iff \neg(\neg c^\dagger) \fatsemi \neg c \leq id^\bullet_Y && \text{(10)}\\
&\iff (\neg\neg c)^\dagger \fatsemi \neg c \leq id^\bullet_Y && \text{(Lemma 3.2)}\\
&\iff c^\dagger \fatsemi \neg c \leq id^\bullet_Y && \text{(Definition 3.1.1)}\\
&\iff (c^\dagger \fatsemi \neg c)^\dagger \leq (id^\bullet_Y)^\dagger && \text{(Table 1)}\\
&\iff \neg c^\dagger \fatsemi c^{\dagger\dagger} \leq id^\bullet_Y && \text{(Table 1)}\\
&\iff \neg c^\dagger \fatsemi c \leq id^\bullet_Y && \text{(Table 1)}
\end{aligned}$$

and similarly for $id^\circ_X \leq c \fatsemi (\neg c)^\dagger$. □

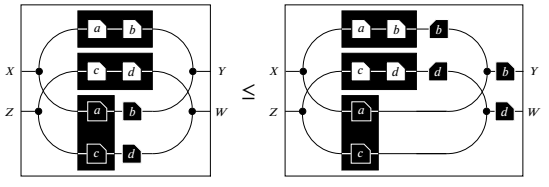
Linear strengths. Now we prove the linear strenghts axioms, namely (ν°_l), (ν°_r), $(\nu^\bullet_l)$, $(\nu^\bullet_r)$, $(\otimes^\bullet)$ and $(\otimes^\circ)$. This yields that every peircean bicategory is a closed symmetric monoidal linear bicategory.

Lemma 3.10. *For any a, b, c, d properly typed, it holds that*

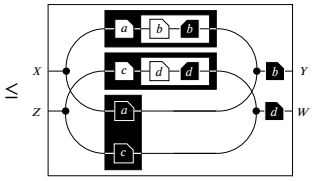


Proof. First observe that the last two inequalities can be proved from the first two, by means of Proposition 3.1.2.

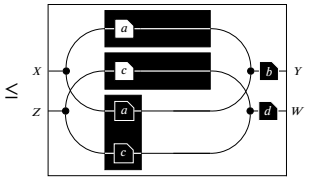
For the first inequality observe that the following holds:



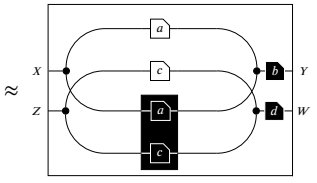
(Lemma 2.1.2)

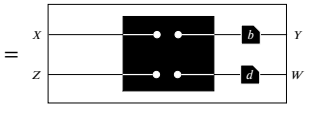


(Lemma 3.8)

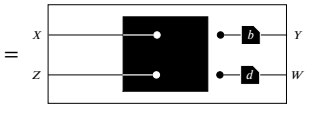


(Lemma 3.9)

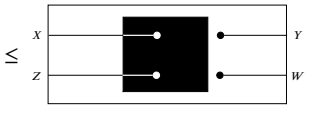




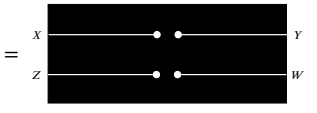
(Definition 3.1.1)



(Lemma 3.4)



(!°-nat)



(Lemma 3.4)

$= \bot_{X \otimes Z, Y \otimes W}$ (Lemma 3.1)

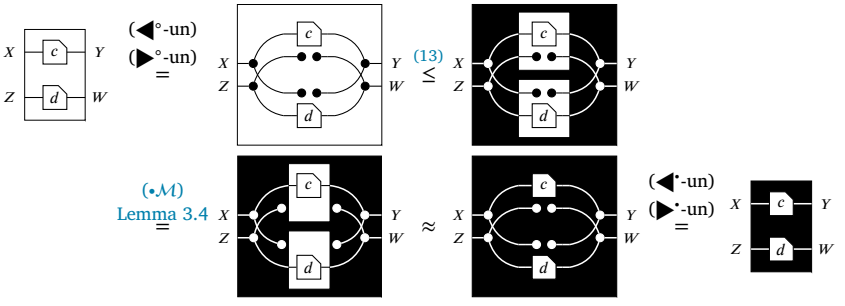
and thus we conclude by Proposition 3.1.4. The proof for the second inequality is analogous. □

Lemma 3.11. *For any $c : X \to Y, d : Z \to W$, it holds that $c \otimes d \leq c \otimes\!\!\!\bullet\, d$.*

Proof. Since $\mathbf{C}[X, Y]$ is a boolean algebra, for any pair of arrows $c, d : X \to Y$ it holds that

$$c \wedge d \leq c \vee d. \tag{13}$$

To conclude observe that the following holds:



□

Axioms $(\otimes\!\!\!\bullet\,\circ)$ and $(\otimes\!\!\!\bullet\,\bullet)$ follow directly from Lemma 3.11 when c and d are identities.

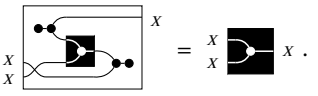
Corollary 3.1. *For all objects X, Y, it holds that*

$$id^{\bullet}_X \otimes id^{\bullet}_Y \leq id^{\bullet}_X \otimes\!\!\!\bullet\, id^{\bullet}_Y \quad \text{and} \quad id^{\circ}_X \otimes id^{\circ}_Y \leq id^{\circ}_X \otimes\!\!\!\bullet\, id^{\circ}_Y.$$

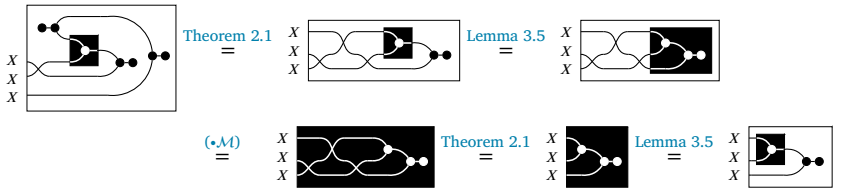
Linear Frobenius. We now arrive at the final step of the main proof: the linear Frobenius axioms—(F$\cdot_\circ$), (F$\circ_\cdot$), (F.$\circ$), and (F$_\circ$.$\cdot$).

First, we prove that the black structure — in particular, the cocopier $\blacktriangleright^\bullet_X$ — can be *rewired* using the white structure.

Proposition 3.3. *For any object X, it holds that*

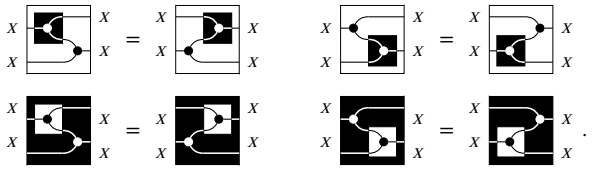


Proof. Observe that the following holds:



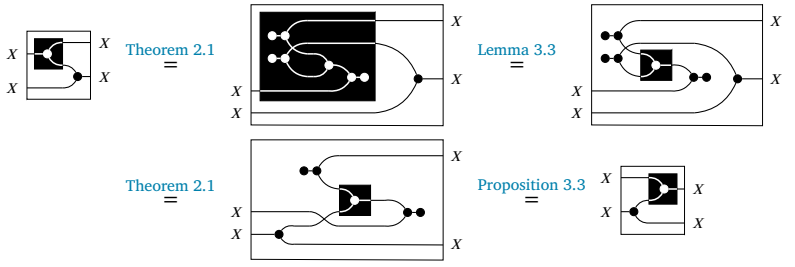
and thus we conclude by Lemma 2.1.1. □

Lemma 3.12. *For any object X, it holds that*



Proof. First observe that the last two equalities can be proved from the first two, by means of Proposition 3.1.2.

For the first equality observe that the following holds:



The second equality is proved analogously. □

PROOF OF THEOREM 3.1. By Propositions 2.4 and 2.6 every fo-bicategory is a peircean bicategory. By Proposition 2.5 every morphism of fo-bicategories is a morphism of peircean bicategories.

Now observe that a peircean bicategory $\mathbf{C}$ is a fo-bicategory, since:

- it is a cartesian bicategory by definition;
- it is a cocartesian bicategory via the isomorphism $\neg : (\mathbf{C}^{co}, \blacktriangleleft^\circ, \blacktriangleright^\circ) \to (\mathbf{C}, \blacktriangleleft^\bullet, \blacktriangleright^\bullet)$;
- it is a closed linear bicategory, since:
  - (δ_l) and (δ_r) hold by Lemma 3.8;
  - (ν_l°), (ν_r°), $(\nu_l^\bullet)$ and $(\nu_r^\bullet)$ hold by Lemma 3.10 and $(\otimes^\bullet)$ and $(\otimes^\circ)$ hold by Corollary 3.1;
  - every arrow $c : X \to Y$, and in particular $\sigma^\circ, \sigma^\bullet, \blacktriangleleft^\circ, !^\circ, \blacktriangleright^\circ, ¡^\circ, \blacktriangleleft^\bullet, !^\bullet, \blacktriangleright^\bullet$ and $¡^\bullet$, has both a left and right linear adjoint by Lemma 3.9;
- (F$\cdot_\circ$), (F$\circ_\cdot$), (F.$\circ$) and (F$_\circ$.$\cdot$) hold by Lemma 3.12;

A morphism of peircean bicategories $F : \mathbf{C} \to \mathbf{D}$ preserves the structure defined in (10) since it preserves negation, e.g.

$$\begin{aligned}
F(c \mathbin{;} d) &= F(\neg(\neg c \mathbin{⨟} \neg d)) && \text{(10)}\\
&= \neg F(\neg c \mathbin{⨟} \neg d) && (F \text{ preserves negation})\\
&= \neg(F(\neg c) \mathbin{⨟} F(\neg d)) && (F \text{ is a strong symmetric monoidal functor})\\
&= \neg(\neg F(c) \mathbin{⨟} \neg F(d)) && (F \text{ preserves negation})\\
&= F(c) \mathbin{;} F(d) && \text{(10)}
\end{aligned}$$

In particular, F is both a morphism of cartesian and cocartesian bicategories and thus a morphism of fo-bicategories. □

4. On elementary-existential doctrines and cartesian bicategories

The notion of hyperdoctrine was introduced by Lawvere in a series of seminal papers [26,28], in order to provide an algebraic framework to represent a first-order theory and its corresponding Tarski semantics. Over the years, various generalisations and specialisations of this concept have been formulated and applied across multiple domains in the fields of logic and computer science. In this section, we recall a generalisation of the notion of hyperdoctrine introduced by Maietti and Rosolini in [47–49], namely that of *elementary and existential doctrine*, and how this notion is related to that of cartesian bicategory [46].

Before proceeding, it is useful to provide some intuition for readers who may be unfamiliar with the underlying ideas. Consider the powerset $\mathcal{P}(X)$ of a set X. We regard its elements as predicates, which we denote by Greek letters $\alpha, \beta, \ldots$. Of particular importance is the predicate $\delta_X \in \mathcal{P}(X \times X)$ defined by

$$\delta_X \stackrel{\text{def}}{=} \{(x_1, x_2) \in X \times X \mid x_1 = x_2\}. \tag{14}$$

Recall that every function $f : X \to Y$ induces a map $\mathcal{P}_f : \mathcal{P}(Y) \to \mathcal{P}(X)$ together with two maps $\exists_f, \forall_f : \mathcal{P}(X) \to \mathcal{P}(Y)$, defined by

$$\begin{aligned} \mathcal{P}_f(\beta) &\stackrel{\text{def}}{=} \{x \in X \mid f(x) \in \beta\}, \\ \exists_f(\alpha) &\stackrel{\text{def}}{=} \{y \in Y \mid \exists x \in X.\, (f(x) = y \wedge x \in \alpha)\}, \\ \forall_f(\alpha) &\stackrel{\text{def}}{=} \{y \in Y \mid \forall x \in X.\, (f(x) = y \Rightarrow x \in \alpha)\}, \end{aligned} \tag{15}$$

for all $\alpha \in \mathcal{P}(X)$ and $\beta \in \mathcal{P}(Y)$. In the special case where f is the projection $\pi_X : X \times Y \to X$, the maps $\exists_{\pi_X}, \forall_{\pi_X} : \mathcal{P}(X \times Y) \to \mathcal{P}(X)$ take the form

$$\exists_{\pi_X}(\alpha) \stackrel{\text{def}}{=} \{x \mid \exists y \in Y.\, (x, y) \in \alpha\} \quad \text{and} \quad \forall_{\pi_X}(\alpha) \stackrel{\text{def}}{=} \{x \mid \forall y \in Y.\, (x, y) \in \alpha\}, \tag{16}$$

thus recovering the familiar semantics of existential and universal quantification. From an algebraic perspective, the crucial observation is that $\exists_f$ is left adjoint to $\mathcal{P}_f$, namely $\alpha \subseteq \mathcal{P}_f \exists_f(\alpha)$ and $\exists_f \mathcal{P}_f(\beta) \subseteq \beta$ for all $\alpha \in \mathcal{P}(X)$ and $\beta \in \mathcal{P}(Y)$. Dually, $\forall_f$ is right adjoint to $\mathcal{P}_f$. Moreover, the assignments $X \mapsto \mathcal{P}(X)$ and $f \mapsto \mathcal{P}_f$ define the well-known *contravariant powerset functor* $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \to \mathsf{Set}$.

Doctrines abstract the structure described above by replacing the category of sets with an arbitrary category $\mathbf{C}$ having finite products, and considering contravariant functors $P : \mathbf{C}^{\mathrm{op}} \to \mathbf{D}$. Different classes of doctrines arise by varying the target category $\mathbf{D}$ among suitable categories of lattice-theoretic structures, such as distributive lattices or Heyting algebras. For example, the Boolean hyperdoctrines recalled in Section 5 are obtained by taking $\mathbf{D} = \mathsf{Bool}$, the category of Boolean algebras and Boolean homomorphisms. Instead, the elementary and existential doctrines –that are the main subject of this section– take values in InfSl, the category of inf-semilattices and inf-preserving monotone maps.

4.1. Elementary and existential doctrines

Before giving the formal definition, it is convenient to introduce terminology and notation that it is often used in the literature: for a category $\mathbf{C}$ with finite products, we write I for a final object and $X \times Y$ for a product of two objects X and Y. As usual, projections are denoted as $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$, the pairing of $f : Z \to X$ and $g : Z \to Y$ as $\langle f, g \rangle : Z \to X \times Y$ and $\Delta_X : X \to X \times X$ stands for $\langle id^\circ_X, id^\circ_X \rangle$. For a functor $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ and an arrow $f : X \to Y$ in $\mathbf{C}$ we write P_f for the morphism of inf-semilattices $P(f) : P(Y) \to P(X)$. The latter is often referred to as a functor.

Definition 4.1. Let $\mathbf{C}$ be a category with finite products, An *elementary and existential doctrine* is a functor $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ satisfying the following properties:

- for every Y in $\mathbf{C}$ there exists an element δ_Y in $P(Y \times Y)$, called *equality predicate*, such that for a morphism $id^\circ_X \times \Delta_Y : X \times Y \to X \times Y \times Y$ in $\mathbf{C}$ and every element α in $P(X \times Y)$, the assignment

$$\exists_{id^\circ_X \times \Delta_Y}(\alpha) \stackrel{\text{def}}{=} P_{\langle \pi_1, \pi_2 \rangle}(\alpha) \wedge P_{\langle \pi_2, \pi_3 \rangle}(\delta_Y)$$

  determines a left adjoint to the functor $P_{id^\circ_X \times \Delta_Y} : P(X \times Y \times Y) \to P(X \times Y)$;
- for any projection $\pi_X : X \times Y \to X$, the functor $P_{\pi_X} : P(X) \to P(X \times Y)$ has a left adjoint $\exists_{\pi_X}$, and these satisfy:

(BCC) *Beck-Chevalley condition:* for any pullback diagram

$$\begin{array}{ccc} X \times Y' & \xrightarrow{\pi_Y} & Y' \\ {\scriptstyle id^\circ_X \times f}\downarrow & & \downarrow{\scriptstyle f} \\ X \times Y & \xrightarrow[\pi_X]{} & Y \end{array}$$

it holds that

$$\exists_{\pi_Y}(P_{id^\circ_X \times f}(\alpha)) = P_f(\exists_{\pi_X}(\alpha))$$

for every element α in $P(X \times Y)$;

(FR) *Frobenius reciprocity:* for every element β in $P(X)$ and α in $P(X \times Y)$, it holds that

$$\exists_{\pi_X}(P_{\pi_X}(\beta) \wedge \alpha) = \beta \wedge \exists_{\pi_X}(\alpha).$$

The archetypical example is provided by the controvariant powerset functor $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ corestricted to InfSl. Indeed, one can easily check that $\mathcal{P}_f$ preserves inf; that the equality predicate in (14) enjoys the first property and that the left adjoint (16) satisfies both (BCC) and (FR).

Example 4.1. More generally, for any regular category $\mathbf{C}$, the subobjects functor $\mathsf{Sub}_{\mathbf{C}} : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ is an elementary and existential doctrine, see [47,48].

Example 4.2. Let $\mathcal{L}_{=,\exists}$ be the $(\top, \wedge, =, \exists)$-fragment of first-order Intuitionistic Logic (for simplicity single sorted) $\mathcal{L}$, called *regular* in [30, Sec. D1.3], i.e. the fragment with the true constant, conjunction, equality and existential quantifiers. Let TH be a theory in such a fragment. Let us denote by $\mathcal{V}$ the syntactic category whose objects are contexts (up to α-equivalence), and arrows are term substitutions. Consider the functor

$$\mathsf{Lt}^{\mathrm{TH}}_{=,\exists} : \mathcal{V}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$$

defined on a given context Γ of $\mathcal{V}$ by taking $\mathsf{Lt}^{\mathrm{TH}}_{=,\exists}(\Gamma)$ as the Lindenbaum-Tarski algebra of well-formed formulas of $\mathcal{L}_{=,\exists}$ with free variables in Γ and on a substitution morphism between contexts by taking the substitution homomorphism between formulas of the Lindenbaum-Tarski algebras. The functor $\mathsf{Lt}^{\mathrm{TH}}_{=,\exists}$ is an elementary and existential doctrine.

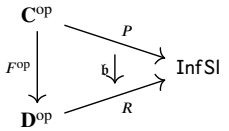
Remark 4.1. In an elementary and existential doctrine, for every arrow $f : X \to Y$ in $\mathbf{C}$, the functor $P_f : P(Y) \to P(X)$ has a left adjoint $\exists_f : P(X) \to P(Y)$ that can be computed for all $\alpha \in P(X)$ as

$$\exists_f(\alpha) \overset{\mathrm{def}}{=} \exists_{\pi_Y}(P_{f \times id_Y^\circ}(\delta_Y) \wedge P_{\pi_X}(\alpha)).$$

These left ajoints satisfy the Frobenius reciprocity but not necessarily the Beck-Chevalley condition. See [60, Rem. 6.4].

Definition 4.2. Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ and $R : \mathbf{D}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be two elementary and existential doctrines. A *morphism of elementary and existential doctrines* is given by a pair $(F, \mathfrak{b})$ where

- $F : \mathbf{C} \to \mathbf{D}$ is a finite product preserving functor;
- $\mathfrak{b} : P \to F^{\mathrm{op}} \circ R$ is a natural transformation;



satisfying the following conditions:

1. for every object X of $\mathbf{C}$, $\mathfrak{b}_{X\times X}(\delta_X) = \delta_{FX\times FX}$;
2. for every $\pi_X : X \times Y \to X$ of $\mathbf{C}$ and for every α in $P(X \times Y)$, it holds that $\exists_{F(\pi_X)}(\mathfrak{b}_{X\times Y}(\alpha)) = \mathfrak{b}_X(\exists_{\pi_X}(\alpha))$.

We write $\mathbb{EED}$ for the category of elementary and existential doctrines and morphisms.

Similarly to cartesian bicategories, elementary and existential doctrines have enough structure to express the notion of *functional* (or single-valued) and *entire* (total) predicates, introduced in [52] as follows.

Definition 4.3. Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. An element $\alpha \in P(X \times Y)$ is said to be *functional from X to Y* if $P_{\langle \pi_1, \pi_2 \rangle}(\alpha) \wedge P_{\langle \pi_1, \pi_3 \rangle}(\alpha) \leq P_{\langle \pi_2, \pi_3 \rangle}(\delta_Y)$ in $P(X \times Y \times Y)$. Also, α is said to be *entire from X to Y* if $\top_X \leq \exists_{\pi_X}(\alpha)$ in $P(X)$.

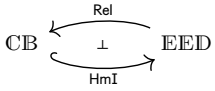
In the doctrine $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$, an element $\alpha \in \mathcal{P}(X \times Y)$ is functional if and only if it defines a partial function from X to Y, while it is entire if it provides a total relation from X to Y.

Lemma 4.1. *A morphism of elementary and existential doctrines preserves both functional and entire elements.*

Proof. It follows from the fact that, by definition, a morphism of elementary and existential doctrines preserves both $\exists_{\pi_X}$ and δ_Y. □

4.2. The adjunction $\mathbb{CB} \dashv \mathbb{EED}$

In [46], it is shown an adjunction between cartesian bicategories and elementary and existential doctrines.

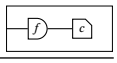
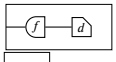


(17)

In this section, we recall the key aspects of such adjunction.

The inclusion $\mathsf{HmI} : \mathbb{CB} \hookrightarrow \mathbb{EED}$:

Table 2
For a cartesian bicategory $\mathbf{C}$, the language of the doctrine $\mathbf{C}[-,I]:\mathsf{Map}(\mathbf{C})^{\mathrm{op}}\longrightarrow\mathsf{InfSl}$ is illustrated above where $f:X\to Y$ is a map of $\mathbf{C}$ while $c:Y\to I$ and $d,c_1,d_1:X\to I$ are arbitrary arrows.

$\mathbf{C}[f,I](c)$	$=$	$f\fatsemi c$	drawn as	
$\exists_f(d)$	$=$	$f^\dagger\fatsemi d$	drawn as	
δ_X	$=$	$\blacktriangleright^\circ_X\fatsemi !^\circ_X$	drawn as	
$\top_X$	$=$	$!^\circ_X$	drawn as	
$c_1\wedge d_1$	$=$	$\blacktriangleleft^\circ_X\fatsemi(c_1\otimes d_1)$	drawn as	

- for a cartesian bicategory $\mathbf{C}$, the doctrine $\mathsf{HmI}(\mathbf{C})$ is given by the hom-functor

  $$\mathbf{C}[-,I]:\mathsf{Map}(\mathbf{C})^{\mathrm{op}}\longrightarrow\mathsf{InfSl}$$

  restricted to $\mathsf{Map}(\mathbf{C})$. Note that the latter is a category with finite products by Proposition 2.1. Moreover, by Proposition 2.2.1 every homset $\mathbf{C}[X,I]$ carries an inf-semillatice. The reader can easily check that, for all maps f, $\mathbf{C}[f,I]$ preserves meets and top by means of the two inequalities in (4). More generally, it holds that $\mathbf{C}[-,I]:\mathsf{Map}(\mathbf{C})^{\mathrm{op}}\longrightarrow\mathsf{InfSl}$ is an elementary and existential doctrine (see [46, Theorem 20]). For convenience of the reader we summarise from [46] the relevant structure of the doctrine $\mathbf{C}[-,I]$ in Table 2.
- a morphism of cartesian bicategories $F:\mathbf{C}\to\mathbf{D}$ is mapped to the morphism of doctrines $\mathsf{HmI}(F)\stackrel{\mathrm{def}}{=}(\tilde{F},\mathfrak{b}^F)$ where $\tilde{F}:\mathsf{Map}(\mathbf{C})\to\mathsf{Map}(\mathbf{D})$ is the functor F restricted to $\mathsf{Map}(\mathbf{C})$ (see Proposition 2.1) and $\mathfrak{b}^F_X:\mathbf{C}[X,I]\to\mathbf{D}[F(X),I]$ is defined, for all objects X of $\mathbf{C}$ and arrows $c\in\mathbf{C}[X,I]$, as

  $$\mathfrak{b}^F_X(c)\stackrel{\mathrm{def}}{=}F(c). \tag{18}$$

Next, we recall the definition of the functor Rel. The main intuition is that such a functor provides a generalisation to elementary and existential doctrines of the construction of bicategory of relations associated with a regular category (see [44, Ex. 1.4]).

The functor $\mathsf{Rel}:\mathbb{EED}\to\mathbb{CB}$:

- For $P:\mathbf{C}^{\mathrm{op}}\longrightarrow\mathsf{InfSl}$, the cartesian bicategory $\mathsf{Rel}(P)$ is defined as follows:
  (a) objects are those of $\mathbf{C}$;
  (b) for objects X,Y, the homsets $\mathsf{Rel}(P)[X,Y]$ are the posets $P(X\times Y)$;
  (c) the identity for an object X is the equality predicate δ_X in $P(X\times X)$;
  (d) composition is defined for all $\phi:X\to Y$ and $\psi:Y\to Z$ as

  $$\phi\fatsemi\psi\stackrel{\mathrm{def}}{=}\exists_{\pi_{X\times Z}}(P_{\pi_{X\times Y}}(\phi)\wedge P_{\pi_{Y\times Z}}(\psi)).$$

- For a morphism of doctrines $(F,\mathfrak{b}):P\to Q$, the morphism of cartesian bicategories $\mathsf{Rel}(F,\mathfrak{b}):\mathsf{Rel}(P)\to\mathsf{Rel}(Q)$ is defined as

  $$\mathsf{Rel}(F,\mathfrak{b})(X)\stackrel{\mathrm{def}}{=}F(X)\qquad\mathsf{Rel}(F,\mathfrak{b})(\phi)\stackrel{\mathrm{def}}{=}\mathfrak{b}_{X\times Y}(\phi) \tag{19}$$

  for all objects X,Y and arrows $\phi:X\to Y$ in $\mathsf{Rel}(P)$.

To see that $\mathsf{Rel}(P)$ is a cartesian bicategory it is convenient to first illustrate the monoidal product. For $\phi:X\to Y$ and $\psi:U\to V$ arrows of $\mathsf{Rel}(P)$, namely $\phi\in P(X\times Y)$ and $\psi\in P(U\times V)$, the arrow $\phi\otimes\psi:X\otimes U\to Y\otimes V$ is defined as

$$\phi\otimes\psi\stackrel{\mathrm{def}}{=}P_{\langle\pi_X,\pi_Y\rangle}(\phi)\wedge P_{\langle\pi_U,\pi_V\rangle}(\psi)$$

where $\langle\pi_X,\pi_Y\rangle$ and $\langle\pi_U,\pi_V\rangle$ are the projections from $X\times U\times Y\times Z$ to, respectively, $X\times Y$ and $U\times V$.

The rest of the structure of cartesian bicategory is inherited from the finite products category $\mathbf{C}$ by means of the *graph functor* $\Gamma_P:\mathbf{C}\to\mathsf{Rel}(P)$. This functor acts as the identity on objects and maps each arrow $f:X\to Y$ in

$$\Gamma_P(f)\stackrel{\mathrm{def}}{=}P_{f\times id_Y}(\delta_Y)\in P(X\times Y)=\mathsf{Rel}(P)[X,Y].$$

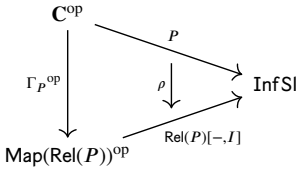
For instance, for the doctrine $\mathcal{P}:\mathsf{Set}^{\mathrm{op}}\longrightarrow\mathsf{InfSl}$, the functor $\Gamma_{\mathcal{P}}:\mathsf{Set}\to\mathbf{Rel}^\circ$ maps every function f to its graph.

At this point, it is worth to observe that the arrow $P_{id_Y\times f}(\delta_Y)$ is right adjoint to $\Gamma_P(f)$ (see [46, Pro. 23]) and thus $\Gamma_P(f)$ is a map. Therefore, Γ_P restricts to a finite-product preserving functor $\mathbf{C}\to\mathsf{Map}(\mathsf{Rel}(P))$. One can thus define the (co)monoid structure of $\mathsf{Rel}(P)$ as

$$\blacktriangleleft^\circ_X\stackrel{\mathrm{def}}{=}\Gamma_P(\blacktriangleleft^\circ_X)\qquad\blacktriangleright^\circ_X\stackrel{\mathrm{def}}{=}P_{id_{X\times X}\times\blacktriangleleft^\circ_X}(\delta_{X\times X})$$
$$!^\circ_X\stackrel{\mathrm{def}}{=}\Gamma_P(!^\circ_X)\qquad ¡^\circ_X\stackrel{\mathrm{def}}{=}P_{id_I\times!^\circ_X}(\delta_I)$$

where on the right hand side of the above equation $\blacktriangleleft^{\circ}_{X} : X \to X \times X$ and $!^{\circ}_{X} : X \to I$ are copier and discard of $\mathbf{C}$ (they exist since $\mathbf{C}$ has finite products).

The graph functor $\Gamma_P : \mathbf{C} \to \mathsf{Map}(\mathsf{Rel}(P))$ is also used for defining the unit of the adjunction (17). For every elementary and existential doctrine $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$, the morphism of elementary and existential doctrines $\eta_P : P \to \mathsf{HmI}(\mathsf{Rel}(P))$ is



(20)

where

- $\Gamma_P : \mathbf{C} \to \mathsf{Map}(\mathsf{Rel}(P))$ is the graph-functor;
- each component $\rho_X : P(X) \to \mathsf{Rel}(P)[X, I] = P(X \times I)$ is given by the isomorphism $P(X) \cong P(X \times I)$ obtained by applying the functor P to the right unitor $X \times I \cong X$ in $\mathbf{C}$.

Hence, the unit of the adjunction $\mathsf{Rel}(-) \dashv \mathsf{HmI}(-)$ is the morphism of elementary and existential doctrines $\eta_P \overset{\text{def}}{=} (\Gamma_P, \rho) : P \to \mathsf{HmI}(\mathsf{Rel}(P))$.

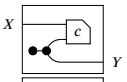
Remark 4.2. Notice that each component of ρ is an iso by definition.

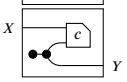
Remark 4.3. Since for any cartesian bicategory $\mathbf{C}$, it holds that $\mathbf{C}[X, Y] \cong \mathbf{C}[X \otimes Y, I]$, one can easily see that $\mathsf{Rel}(\mathsf{HmI}(\mathbf{C})) \equiv \mathbf{C}$, i.e. the counit of the adjunction is an iso.

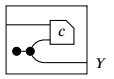
4.3. On maps and functional and entire elements

Hereafter we show a result (Proposition 4.1) about the adjunction (17) that is crucial for Section 5 but it is not proved in [46]. We start with the following lemma that provides a precise correspondence between the notion of map and that of functional and entire element.

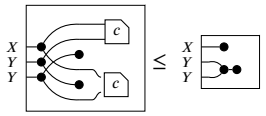
Lemma 4.2. *Let $\mathbf{C}$ be a cartesian bicategory and $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ the corresponding doctrine. An element $c \in \mathbf{C}[X \times Y, I]$ is*

functional in $\mathbf{C}[-, I]$ iff  *is single valued in $\mathbf{C}$;*

entire in $\mathbf{C}[-, I]$ iff  *is total in $\mathbf{C}$.*

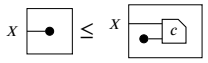
Therefore, c is both functional and entire if and only if  *is a map.*

Proof. By definition, an element $c \in \mathbf{C}[X \times Y, I]$ is functional from X to Y if and only if



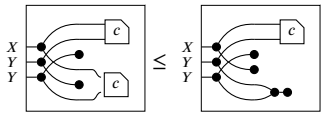
(functional)

and it is entire from X to Y if and only if



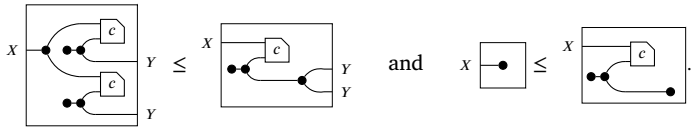
(entire)

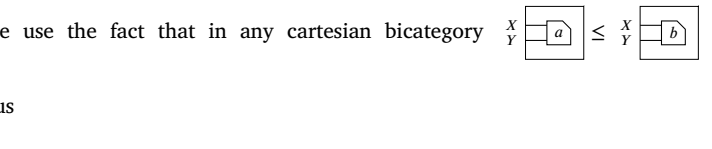
Now observe that (functional) holds if and only if the following holds

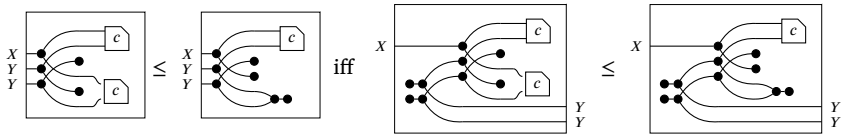


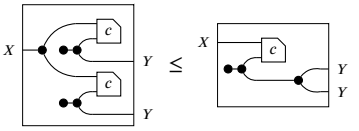
(functional-alt)

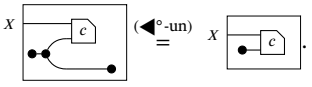
To conclude, we need to show that (functional-alt) and (entire) are equivalent, respectively, to



For (functional-alt) we use the fact that in any cartesian bicategory  if and only if , and thus



which amounts to  using Theorem 2.1.

For (entire) it suffices to observe that  □

By using the above lemma, one can exploit the unit of the adjunction $\mathsf{Rel}(-) \dashv \mathsf{HmI}(-)$ to provide a simple characterisation of the maps of $\mathsf{Rel}(P)$ in terms of functional and entire elements of P.

Proposition 4.1. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. Then the maps of $\mathsf{Rel}(P)$ are precisely the functional and entire elements of P.*

Proof. For every elementary and existential doctrine P, the unit of the adjunction $\mathsf{Rel}(-) \dashv \mathsf{HmI}(-)$ is morphism of elementary and existential doctrines $\eta_P \overset{\text{def}}{=} (\Gamma_P, \rho) : P \to \mathsf{HmI}(\mathsf{Rel}(P))$, and each component of ρ is an iso by definition, see Remark 4.2. Therefore, ρ preserves and reflects functional and entire relations (see Lemma 4.1), i.e. $\rho_{X\times Y}(\phi) \in \mathsf{Rel}(P)[X \times Y, I]$ is functional and entire in $\mathsf{Rel}(P)$ if and only if ϕ is functional and entire in P. On the other hand, by Lemma 4.2 we have that an arrow $\psi \in \mathsf{Rel}(P)[X, Y]$, i.e. an element $\psi \in P(X \times Y)$, is a map if and only if the corresponding element in $\mathsf{Rel}(P)[X \times Y, I]$, i.e. $\rho_{X\times Y}(\psi)$, is functional and entire with respect to the doctrine $\mathsf{Rel}(P)[-, I]$. Therefore, we can conclude that ϕ is a map in $\mathsf{Rel}(P)$ if and only if ϕ is an entire and functional element of P. □

4.4. The equivalence $\mathbb{CB} \equiv \overline{\mathbb{EED}}$

Recall from [46] that the adjunction in (17) becomes an equivalence for certain well-behaved doctrines. The main intuition is that these doctrines are precisely those whose graph functor

$$\Gamma_P : \mathbf{C} \to \mathsf{Map}(\mathsf{Rel}(P))$$

happens to be an isomorphism, i.e. those doctrines where every functional and entire relation arises as graph of an arrow of the base category.

To properly formalize this idea, we recall from [48,52] the notion of doctrine *with comprehensive diagonals* and of doctrine satisfying the *rule of unique choice*. Having comprehensive diagonals intuively means that the graph functor is faithful, while satisfying rule of unique choice means that it is full.

Definition 4.4. An elementary and existential doctrine $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ has *comprehensive diagonals* if for the equality predicate $\delta_X \in P(X \times X)$ it holds that $P_{\Delta_X}(\delta_X) = \top_X$ and every arrow $f : Y \to X \times X$ such that $P_f(\delta_X) = \top_Y$ factors (uniquely) through Δ_X .

Intuitively, a doctrine has comprehensive diagonals if its equality is *extensional*, namely if a formula $t_1 = t_2$ is true, then the terms t_1 and t_2 are syntactically equal. In the language of cartesian bicategories, for two maps t_1, t_2, this can be stated as in Proposition 2.3. While it is sometimes meaningful to consider syntactic doctrines (e.g. Example 4.2) in which the equality is not extensional, in several semantical doctrines this condition is satisfied.

Definition 4.5. Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary existential doctrine. We say that P satisfies the *Rule of Unique Choice* (RUC) if for every entire functional element ϕ in $P(X \times Y)$ there exists an arrow $f : X \to Y$ such that $\top_X \leq P_{\langle id^\circ_X, f\rangle}(\phi)$.

Again, the reader can think that a doctrine has (RUC) if for every element (intuitively formula) that is entire and functional, there exists an arrow in $\mathbf{C}$ (intuitively a term) that represents it.

The doctrine $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ has comprehensive diagonals, and it satisfies the Rule of Unique Choice (RUC) since every functional and total relation can be represented by a function. More generally, we have that:

Example 4.3. Every subobject doctrine $\mathsf{Sub}_{\mathbf{C}} : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ on a regular category, as presented in Example 4.1 satisfies the (RUC) and it has comprehensive diagonals, as observed in [52].

Example 4.4. The doctrine $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ associated with a catesian bicategory $\mathbf{C}$ satisfies the (RUC) and it has comprehensive diagonals, as proved in [46].

Hereafter we denote by $\overline{\mathbb{EED}}$ the full subcategory of $\mathbb{EED}$ whose objects are doctrines satisfying (RUC) and with comprehensive diagonals. We recall the following result [46, Theorem 35]:

Theorem 4.1. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. Then the following two conditions are equivalent:*

- *P has comprehensive diagonals and satisfies (RUC);*
- *$\mathbf{C} \cong \mathsf{Map}(\mathsf{Rel}(P))$ and $P \cong \mathsf{Rel}(P)[-, I]$.*

In particular, the adjunction $\mathsf{Rel}(-) \dashv \mathsf{HmI}(-)$ restricts to an equivalence $\mathbb{CB} \equiv \overline{\mathbb{EED}}$.

5. On boolean hyperdoctrines and first-order bicategories

In the previous section we have recalled from [46] the correspondences between cartesian bicategories and elementary existential doctrines. In this section, we show that the same correspondences hold for fo-bicategories and *boolean hyperdoctrines*. Crucially our proofs rely on the characterisation of fo-bicategories as peircean bicategories of Theorem 3.1.

5.1. Boolean hyperdoctrines

In this section we recall the notion of boolean hyperdoctrine, and we illustrate some useful properties. Hereafter we write Bool for the category of boolean algebras and their morphisms.

Definition 5.1 (boolean hyperdoctrine)**.** Let $\mathbf{C}$ be a category with finite products. A functor $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ is a *boolean hyperdoctrine* if it is an elementary and existential doctrine. A morphism $(F, \mathfrak{b}) : P \to R$ of boolean hyperdoctrines is a morphism of elementary and existential doctrines such that $\mathfrak{b}_X$ is a morphism of boolean algebras for all objects X of $\mathbf{C}$. We denote by $\mathbb{BHD}$ the category of boolean hyperdoctrines and their morphisms.

The archetypical example of a boolean hyperdoctrine is, again, provided by the controvariant powerset functor $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ coristricted to Bool.

Example 5.1. This can be generalized to an arbitrary *boolean category* $\mathbf{B}$, namely a coherent category such that every subobject has a complement, see [30, Sec. A1.4, p. 38]. The subobjects functor on $\mathbf{B}$ is a boolean hyperdoctrine $\mathsf{Sub}_{\mathbf{B}} : \mathbf{B}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$.

Example 5.2. Given a standard first-order theory TH in a first-order language $\mathcal{L}$ (for simplicity single sorted), one can consider the functor $\mathsf{Lt}^{\mathrm{TH}} : \mathcal{V}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ defined as in Example 4.2. In this case, the functor $\mathsf{Lt}^{\mathrm{TH}}$ is a boolean hyperdoctrine, because we are considering the *all* the formulas of first-order language $\mathcal{L}$, and hence the Lindenbaum-Tarski algebra of well-formed formulas is a boolean algebra.

It is well-known that in first-order logic the universal quantifier can be derived by the existential quantifier and the negation. The same happens in boolean hyperdoctrines: for all $f : X \to Y$, the functor $\forall_f(-) : P(X) \to P(Y)$ is defined for all $\alpha \in P(X)$ as

$$\forall_f(\alpha) \stackrel{\mathrm{def}}{=} \neg\exists_f\neg(\alpha).$$

Lemma 5.1. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ be a boolean hyperdoctrine. Then for every arrow $f : A \to B$, the functor $\forall_f : P(A) \to P(B)$ provides a right adjoint to $P_f : P(B) \to P(A)$. Moreover, if $\exists_f$ satisfies BCC then also $\forall_f$ satisfies BCC.*

Proof. The proof is a straightforward generalisation of the ordinary proof in first-order classical logic. Indeed:

$$\alpha \leq \forall_f(P_f(\alpha)) = \neg\exists_f(\neg P_f(\alpha))$$

holds if and only if

$$\exists_f(\neg P_f(\alpha)) \leq \neg\alpha$$

because P is boolean. But this holds because it is equivalent to

$$P_f(\neg\alpha) \leq P_f(\neg\alpha).$$

To prove that

$$P_f(\forall_f(\beta)) \leq \beta$$

we have to use again the assumption that P is boolean. In fact we have that

$$P_f(\forall_f(\beta)) = \neg P_f(\exists_f(\neg\beta)) \leq \neg\neg\beta$$

because $\neg\beta \leq P_f(\exists_f(\neg\beta))$ (since $\exists_f \dashv P_f$), and using the fact that $\neg\neg\beta = \beta$ we can conclude that $P_f(\forall_f(\beta)) \leq \beta$. Now we prove that if $\exists_f$ satisfies BCC then also $\forall_f$ satisfies BCC. So let us consider a pullback

$$\begin{array}{ccc} D & \xrightarrow{f'} & C \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ A & \xrightarrow[f]{} & B \end{array}$$

and suppose that $P_g(\exists_f(\alpha)) = \exists_{f'}(P_{g'}(\alpha))$ for every α in $P(A)$. From this we can deduce that $\neg P_g(\exists_f(\alpha)) = P_g(\neg\exists_f(\alpha)) = \neg\exists_{f'}(P_{g'}(\alpha))$, and then that $P_g(\forall_f(\neg\alpha)) = \forall_{f'}(P_{g'}(\neg\alpha))$ for every α element of $P(A)$. Therefore, in particular it holds for the element $\beta := \neg\alpha$, and hence (using the fact that P is boolean) we can conclude that $P_g(\forall_f(\alpha)) = \forall_{f'}(P_{g'}(\alpha))$. □

Remark 5.1 (Frobenius reciprocity)**.** By Remark 4.1, in an elementary and existential doctrine, left adjoints $\exists$ exist and satisfy the Frobenius reciprocity (FR) along every morphism of the base category, but it is not guarantee that they satisfy the Beck-Chevalley conditions with respect to pullbacks along an arbitrary morphism. In a boolean hyperdoctrine $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ right adjoints $\forall$ exists along any morphism of the base category by Lemma 5.1 and, employing the preservation of the implication $\to$ by the functor P_f, it is straightforward to check that right adjoints satisfy the Frobenius reciprocity (FR) along every morphism of the base category, namely, for every morphism $f : X \to Y$, α in $P(Y)$ and β in $P(X)$,

$$\forall_f(P_f(\alpha) \to \beta) = \alpha \to \forall_f(\beta).$$

See [61, Rem. 1.3]. Moreover, by Lemma 5.1, we have that if $\exists_f$ satisfies Beck-Chevalley condition, then also $\forall_f$ satisfies Beck-Chevalley condition. But, as in the case of left adjoints $\exists$, it is not guaranteed that $\forall$ satisfies Beck-Chevalley conditions with respect to pullbacks along an arbitrary morphism.

We conclude this section with a key result that, intuitively, is the analogous of Proposition 2.6.

Lemma 5.2. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ be a boolean hyperdoctrine and $\phi \in P(X \times Y)$ a functional and entire element from X to$Y$. For all $\psi \in P(Y \times Z)$, it holds that*

$$\exists_{\pi_{X\times Z}}(P_{\pi_{X\times Y}}(\phi) \wedge P_{\pi_{Y\times Z}}(\neg\psi)) = \neg(\exists_{\pi_{X\times Z}}(P_{\pi_{X\times Y}}(\phi) \wedge P_{\pi_{Y\times Z}}(\psi))).$$

Proof. The proof happens to be straightforward if we employ the classical arguments of natural deduction. Here we provide a completely algebric proof. For sake of readability, here we employ the notation given by the internal language of P, writing $\exists y$ instead of the left adjoint $\exists_{\pi_Y}$ and using the predicates $\phi(x, y)$ and $\psi(y, z)$ to denote the elements $\phi \in P(X \times Y)$ and $\psi \in P(Y \times Z)$. Given the well-established correspondence between a doctrine and its internal language, proving that the two previous predicates are equivalent using the logical rules of first-order classical logic is equivalent to prove that they are equal in P. Hereafter we thus prove

$$\exists y.(\phi(x, y) \wedge \neg\psi(y, z)) = \neg\exists y'.(\phi(x, y') \wedge \psi(y', z))$$

Therefore, we start by proving

$$\exists y'.(\phi(x, y') \wedge \neg\psi(y', z)) \vdash \neg\exists y.(\phi(x, y) \wedge \psi(y, z)) \tag{21}$$

The crucial point to conclude that (21) holds is to show that

$$\phi(x, y) \wedge \psi(y, z) \vdash \neg\phi(x, y') \vee \psi(y', z). \tag{22}$$

holds. In fact, from (22) we can deduce the validity of (21) because (22) implies that

$$\exists y.(\phi(x, y) \wedge \psi(y, z)) \vdash \neg\phi(x, y') \vee \psi(y', z)$$

and hence, by applying to both sides $\neg$, we obtain

$$\phi(x, y') \wedge \neg\psi(y', z) \vdash \neg\exists y.(\phi(x, y) \wedge \psi(y, z))$$

and hence that

$$\exists y'.(\phi(x, y') \wedge \neg\psi(y', z)) \vdash \neg\exists y.(\phi(x, y) \wedge \psi(y, z))$$

So we have to prove the validity of (22). Now since we are working with boolean algebras, we have that $\phi(x, y) \wedge \psi(y, z) = \phi(x, y) \wedge \psi(y, z) \wedge (y = y' \vee y \neq y')$. Therefore, (22) is equivalent to

$$(\phi(x, y) \wedge \psi(y, z) \wedge y = y') \vee (\phi(x, y) \wedge \psi(y, z) \wedge y \neq y') \vdash \neg\phi(x, y') \vee \psi(y', z). \tag{23}$$

To prove (23), it is enough to prove that both

$$(\phi(x, y) \wedge \psi(y, z) \wedge y = y') \vdash \neg\phi(x, y') \vee \psi(y', z) \tag{24}$$

and

$$(\phi(x, y) \wedge \psi(y, z) \wedge y \neq y') \vdash \neg\phi(x, y') \vee \psi(y', z) \tag{25}$$

hold. Now notice that (25) holds trivially, because $\phi(x,y) \wedge \psi(y,z) \wedge y = y' \vdash \psi(y',z)$. To prove (25) we have to employ the functionality of ϕ. In fact, by definition of functionality we have that

$$\phi(x,y) \wedge \phi(x,y') \vdash y = y'$$

and then, using the fact that $y = y'$ is equivalent to $(y \neq y') \to \bot$ (were $y \neq y'$ denotes $\neg(y = y')$) we can conclude that

$$\phi(x,y) \wedge y \neq y' \vdash \neg\phi(x,y') \tag{26}$$

and hence that $\phi(x,y) \wedge y \neq y' \vdash \neg\phi(x,y') \vee \psi(y',z)$. Therefore, since $\phi(x,y) \wedge \psi(y,z) \wedge y \neq y' \vdash \phi(x,y) \wedge y \neq y'$, we can conclude by transitivity that also (25) holds.

This concludes the proof that (22), and hence, (21) hold.

Now we have to prove that

$$\neg\exists y.(\phi(x,y) \wedge \psi(y,z)) \vdash \exists y'.(\phi(x,y') \wedge \neg\psi(y',z)) \tag{27}$$

First, notice that (27) is equivalent to

$$\forall y'.(\neg\phi(x,y') \vee \psi(y',z)) \vdash \exists y.(\phi(x,y) \wedge \psi(y,z)). \tag{28}$$

Now, in order to prove (28), we start by employing the assumption that ϕ is total, namely:

$$\top \vdash \exists y'.\phi(x,y'). \tag{29}$$

In fact, (29) implies that

$$\forall y'.\neg\phi(x,y') \vdash \bot$$

and since in every boolean algebra we have that $\psi(y',z) \wedge \neg\psi(y',z) \vdash \bot$, we have that

$$\forall y'.\neg\phi(x,y') = \forall y'.(\neg\phi(x,y') \vee \bot) = \forall y'.(\neg\phi(x,y') \vee (\psi(y',z) \wedge \neg\psi(y',z))) \vdash \bot \tag{30}$$

From (30), using the distributivity of $\vee$, and the fact that the universal quantifier of a disjunction is equivalent to the disjunction of two universal quantifiers (categorically, right adjoints preserve coproducts), we can conclude that

$$\forall y'.(\neg\phi(x,y') \vee \psi(y',z)) \wedge \forall y.(\neg\phi(x,y) \wedge \neg\psi(y,z)) \vdash \bot$$

and hence that

$$\forall y'.(\neg\phi(x,y') \vee \psi(y',z)) \vdash \neg\forall y.(\neg\phi(x,y) \wedge \neg\psi(y,z)).$$

Now, since $\neg\forall y.(\neg\phi(x,y) \wedge \neg\psi(y,z)) = \exists y.(\phi(x,y) \wedge \psi(y,z))$ we can conclude that (28), and then (27) hold. □

5.2. Restricting the adjunction

In this section, we show that the adjunction presented in (17) restricts to an adjunction between peircean bicategories and boolean hyperdoctines.

$$\begin{array}{ccc} \mathbb{PB} & \underset{\bot}{\rightleftarrows} & \mathbb{BHD} \\ \downarrow & & \downarrow \\ \mathbb{CB} & \underset{\bot}{\rightleftarrows} & \mathbb{EED} \end{array} \tag{31}$$

Recall the functor $\mathsf{HmI} : \mathbb{CB} \to \mathbb{EED}$ from Section 4.2. The following results ensures that it maps peircean bicategories into boolean hyperdoctrines.

Proposition 5.1. *Let* $\mathbf{C}$ *be a peircean bicategory. Then* $\mathsf{HmI}(\mathbf{C})$ *is a boolean hyperdoctrine.*

Proof. By (17), $\mathbf{C}[-,I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ is an elementary and existential doctrine and, by definition of peircean bicategories, $\mathbf{C}[X,I]$ is a boolean algebra for all objects X. To conclude that $\mathbf{C}[-,I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{Bool}$, one has only to show that, for all maps $f : X \to Y$, $\mathbf{C}[f,I] : \mathbf{C}[Y,I] \to \mathbf{C}[X,I]$ is a morphism of boolean algebras. Since, by (17), $\mathbf{C}[f,I]$ is a morphism of inf-semilattices, it is enough to show that it preserves negation: for all $c \in \mathbf{C}[Y,I]$

$$\begin{aligned} \mathbf{C}[f,I](\neg c) &= f \fatsemi \neg c && \text{(Definition of } \mathbf{C}[-,I]) \\ &= \neg(f \fatsemi c) && (\neg\mathcal{M}) \\ &= \neg\mathbf{C}[f,I](c) && \text{(Definition of } \mathbf{C}[-,I]) \end{aligned}$$

□

The above proposition allows us to characterize peircean bicategories as follows:

Corollary 5.1. *Let* $\mathbf{C}$ *be a cartesian bicategory. Then it is a peircean bicategory if and only if* $\mathsf{HmI}(\mathbf{C})$ *is a boolean hyperdoctrine.*

Now, recall from Section 4.2 the functor $\mathsf{Rel} : \mathbb{EED} \to \mathbb{CB}$. The following results ensures that it maps boolean hyperdoctrines into peircean bicategories.

Proposition 5.2. *Let P be a boolean hyperdoctrine. Then $\mathsf{Rel}(P)$ is a peircean bicategory.*

Proof. By (17), $\mathsf{Rel}(P)$ is a cartesian bicategory. Since $P(X)$ is a boolean algebra for all objects X, then each hom-set $\mathsf{Rel}(P)[X, Y]$ – by definition $P(X \times Y)$ – is a boolean algebra. To conclude that $\mathsf{Rel}(P)$ is a peircean bicategory, it is enough to show that $(\neg\mathcal{M})$ holds, that is

$$\phi \fatsemi \neg\psi = \neg(\phi \fatsemi \psi)$$

for all maps $\phi \in \mathsf{Rel}(P)[X, Y]$ and arrows $\psi \in \mathsf{Rel}(P)[Y, Z]$. By Proposition 4.1, ϕ is a functional and entire element of P. Thus, one can rely on Lemma 5.2 to conclude that

$$\begin{aligned}
\phi \fatsemi \neg\psi &= \exists_{\pi_{X\times Z}}(P_{\pi_{X\times Y}}(\phi) \wedge P_{\pi_{Y\times Z}}(\neg\psi)) && \text{(Defintion of } \mathsf{Rel}(P)\text{)}\\
&= \neg(\exists_{\pi_{X\times Z}}(P_{\pi_{X\times Y}}(\phi) \wedge P_{\pi_{Y\times Z}}(\psi)))\ && \text{(Lemma 5.2)}\\
&= \neg(\phi \fatsemi \psi) && \text{(Defintion of } \mathsf{Rel}(P)\text{)}
\end{aligned}$$

□

Thanks to Propositions 5.1 and 5.2 proving the following result amounts to a few routine checks.

Theorem 5.1. *The adjunction in (17), restricts to the following adjunction.*

$$\mathbb{PB} \underset{\mathsf{HmI}}{\overset{\mathsf{Rel}}{\underset{\longrightarrow}{\overset{\longleftarrow}{\bot}}}} \mathbb{BHD}.$$

Proof. First, we want to prove that the inclusion $\mathsf{HmI} : \mathbb{CB} \hookrightarrow \mathbb{EED}$ in (17) restricts to an inclusion of categories $\mathbb{PB} \hookrightarrow \mathbb{BHD}$. By Proposition 5.1, one only needs to check for morphisms in $\mathbb{PB}$. Given a morphism of peircean bicategories $F : \mathbf{C} \to \mathbf{D}$, $\mathsf{HmI}(F)$ is the morphism of elementary and existential doctrines $(\tilde{F}, \mathfrak{b}^F)$. In order to conclude that it is a morphism of boolean doctrines, it is enough to show that $\mathfrak{b}^F_X$ is a morphism of boolean algebras for all objects X. Since $(\tilde{F}, \mathfrak{b}^F)$ is a morphism of doctrines, $\mathfrak{b}^F_X$ is a morphism of inf-semilattices. Thus it is enough to show that $\mathfrak{b}^F_X$ preserve negation. But this easily follows from the definition of $\mathfrak{b}^F$ in (18): for all $c \in \mathbf{C}[X, I]$,

$$\begin{aligned}
\mathfrak{b}^F_X(\neg c) &= F(\neg c) && \text{(18)}\\
&= \neg F(c) && \text{(morphism of } \mathbb{PB}\text{, Definition 3.1)}\\
&= \neg\mathfrak{b}^F_X(c) && \text{(18)}
\end{aligned}$$

Now, to prove that Rel restrict to a functor $\mathsf{Rel} : \mathbb{BHD} \to \mathbb{PB}$, by Proposition 5.2, one only needs to check that for all morphisms of boolean hyperdoctrines $(F, \mathfrak{b}) : P \to Q$, $\mathsf{Rel}(F, \mathfrak{b}) : \mathsf{Rel}(P) \to \mathsf{Rel}(Q)$ is a morphism of peicean bicategories. Since by (17), $\mathsf{Rel}(F, \mathfrak{b})$ is a morphism of cartesian bicategories, one only needs to check that it preserves the negation. But this is obvious since for all arrows $\phi \in \mathsf{Rel}(P)[X, Y]$, $\mathsf{Rel}(F, \mathfrak{b})(\phi)$ is –by definition– $\mathfrak{b}_{X\times Y}(\phi)$ and $\mathfrak{b}_{X\times Y}$ is a morphism of boolean algebras.

To conclude, one only needs to check the unit and the counit of the adjunction in (17). The counit is an isomorphism of cartesian bicategories (see Equation (19) in [46]), and then it provides an isomorphism of peircean bicategories $\mathbf{C} \cong \mathsf{Rel}(\mathbf{C}[-, I])$ whenever $\mathbf{C}$ is a peircean bicategory. The unit of the adjunction $\eta_P : P \to \mathsf{Rel}(P)[-, I]$ is the morphism of elementary and existential doctrines (Γ_P, ρ) illustrated in (20). To conclude that η_P is a morphism of boolean hyperdoctrine whenever P is a boolean hyperdoctrine, one has only to prove that ρ is a morphism of boolean algebras, but this is trivial since ρ is always an isomorphism of inf-semilattices (see Remark 4.2).

□

By composing the adjunction above with the equivalence illustrated in Theorem 3.1, we obtain the claimed adjunction between fo-bicategories and boolean hyperdoctrines.

Corollary 5.2. *There is an adjunction*

$$\mathbb{FOB} \underset{\longrightarrow}{\overset{\longleftarrow}{\bot}} \mathbb{BHD}.$$

5.3. Restricting the equivalence

By means of Theorem 5.1, it is easy to prove that the equivalence in Theorem 4.1 restricts to an equivalence between $\mathbb{FOB}$ and $\overline{\mathbb{BHD}}$, the full subcategory of $\mathbb{BHD}$ whose objects are boolean hyperdoctrines satisfying (RUC) and with comprehensive diagonals.

Theorem 5.2. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. Then the following two conditions are equivalent:*

- *P is a boolean hyperdoctrine, it has comprehensive diagonals and satisfies (RUC);*
- *$\mathsf{Rel}(P)$ is a peircean bicategory, $\mathbf{C} \cong \mathsf{Map}(\mathsf{Rel}(P))$ and $P \cong \mathsf{Rel}(P)[-, I]$.*

In particular, we have an equivalence $\mathbb{PB} \equiv \overline{\mathbb{BHD}}$ *and thus* $\mathbb{FOB} \equiv \overline{\mathbb{BHD}}$.

Proof. It follows by combining Theorem 5.1 with Theorem 4.1. In particular, if P is a boolean hyperdoctrine, it has comprehensive diagonals and satisfies (RUC) then, by Theorem 4.1, $\mathbf{C} \cong \mathsf{Map}(\mathsf{Rel}(P))$ and $P \cong \mathsf{Rel}(P)[-, I]$. Moreover, since P is a boolean hyperdoctrine, by Theorem 5.1, $\mathsf{Rel}(P)$ is a peircean bicategory. On the other hand, if $\mathsf{Rel}(P)$ is a peircean bicategory, $\mathbf{C} \cong \mathsf{Map}(\mathsf{Rel}(P))$ and $P \cong \mathbf{Rel}(P)[-, I]$, then P is a boolean hyeprdoctine because, by Theroem 5.1, the doctrine $\mathbf{Rel}(P)[-, I]$ associated with a peircean bicategory. Finally, p has comprehensive diagonals and satisfies (RUC) by Theroem 4.1. □

6. Comparing boolean categories with first-order bicategories

The main purpose of this section is to show the equivalence

$$\mathbb{BC} \equiv \mathbb{FOB}_f$$

between the category $\mathbb{BC}$ of *boolean categories*, i.e. coherent categories such that every subobject has a complement, see [30, Sec. A1.4, p. 38] (and also Example 5.1), and the category $\mathbb{FOB}_f$ of those fo-bicategories that are *functionally complete* in the sense of Carboni and Walters [44].

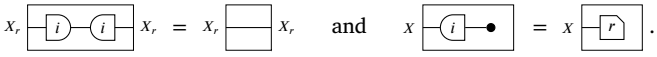
In order to prove this result, we will employ the theory of boolean hyperdoctrines, and in particular the notion of *full comprehension* introduced in the context of doctrines in [47] as a bridge between the two sides of the equivalence.

The key point will be showing that the concept of full comprehension and that of tabulation are equivalent when we consider the doctrine associated with a cartesian bicategory.

6.1. Functionally complete cartesian bicategories

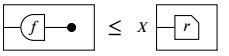
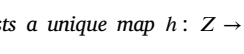
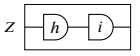
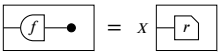
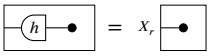
Functionally complete cartesian bicategories were introduced in [44], and they essentially are cartesian bicategories where we can formally represent the concept of *domain of definition of a relation*. In detail:

Definition 6.1. A cartesian bicategory is *functionally complete* if for every arrow $r : X \to I$ there exists an object X_r and a map $i : X_r \to X$, called *tabulation of* r, such that $i \fatsemi i^\dagger = id^\circ_{X_r}$ and $i^\dagger \fatsemi !^\circ_{X_r} = r$. In diagrams,



As expected, we say that a fo-bicategory is functionally complete if its white structure is functionally complete. We write $\mathbb{CB}_f$ for the full subcategory of $\mathbb{CB}$ given by functionally complete cartesian bicategories and, similarly, $\mathbb{FOB}_f$ for functionally complete fo-bicategories.

Lemma 6.1. *Let* $i : X_r \to X$ *be a tabulation of* $r : X \to I$. *Then*

- *for every map* $f : Z \to X$ *such that*  *there exists a unique map* $h : Z \to X_r$ *such that*  
- *additionally, if*  *, then* 

Proof. See [44, Lem. 3.3]. □

In the cartesian bicategory **Rel**°, the tabulation of a relation $r : X \to I$ is given by the subset $X_r \subseteq X$ of those elements of X on which r is defined, together with the trivial inclusion $i : X_r \hookrightarrow X$. In this setting, Lemma 6.1 simply amounts to the familiar fact that, for every function $f : Z \to X$, if its image is included in X_r, then it factors uniquely as $h \fatsemi i$. Moreover, if the image of f coincides with X_r, then h is surjective.

6.2. Doctrines with full comprehensions

The essential intuition behind the concept of tabulation is that a tabulation represents the domain of definition of a relation. A notion aiming to abstract the same concept has been introduced in the context of fibrations in [51] and, as particular instance, in the context of doctrines in [47], under the name of *comprehensions*.

Definition 6.2. Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathtt{InfSl}$ be an elementary and existential doctrine, X an object of $\mathbf{C}$ and α an element of $P(X)$. A *comprehension* of α is an arrow $\{\alpha\} : X_\alpha \to X$ such that

- $P_{\{\alpha\}}(\alpha) = \top_{X_\alpha}$ and,
- for every $f : Y \to X$ such that $P_f(\alpha) = \top_Y$, there exists a unique arrow $g : Y \to X_\alpha$ such that $f = g \fatsemi \{\alpha\}$.

We say that P *has comprehensions* if every α has a comprehension. We say that P has *full comprehensions* if it has comprehensions and, $\alpha \leq \beta$ whenever $\{\alpha\}$ factors through $\{\beta\}$.

We write $\overline{\mathbb{EED}}_c$ and $\overline{\mathbb{BHD}}_c$ for the full subcategory of $\mathbb{EED}$, respectively $\mathbb{BHD}$, of those doctrines with full comprehension, comprehensive diagonals and satisfying (RUC).

The boolean hyperdoctrine $\mathcal{P} : \mathsf{Set}^{\mathrm{op}} \longrightarrow \mathsf{Bool}$ has full comprehensions. In particular the comprehension of a subset $A \subseteq X$ is given by the inclusion $\{A\} : A \to X$ of A in X. More generally, we have the following example:

Example 6.1. Every subobject doctrine $\mathsf{Sub}_{\mathbf{C}} : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ on a regular category (see EXample 4.1) has full comprehensions, as observed in [52]. In this case, the comprehension of a subobject is given by the subobject itself.

Remark 6.1. In the light of Definition 6.2, we can rephrase Definition 4.4, saying that an elementary doctrine has comprehensive diagonals if Δ_X is the comprehension of δ_X.

We recall here some useful properties of comprehensions.

Lemma 6.2. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elemantary existential doctrine with full comprehensions and comprehensive diagonals. Let α be an element of $P(X)$ and $\{\alpha\} : X_\alpha \to X$ be its comprehension. Then*

1. $\alpha = \exists_{\{\alpha\}}(\top_{X_\alpha})$;
2. $\{\alpha\} : X_\alpha \to X$ *is a mono;*
3. $P_{\{\alpha\}\times\{\alpha\}}(\delta_X) = \delta_{X_\alpha}$

Proof. For the first two points, see [60, Prop. 7.10]. For the third point we exploit [62, Prop. 2.33]: arrows in $\mathbf{C}$ are mono iff they are injective in the logic of the functor; namely

$$f : X \to Y \text{ is mono if and only if } P_{f\times f}(\delta_Y) = \delta_X.$$

Thus, replacing f by $\{\alpha\}$ and using the fact that $\{\alpha\}$ is mono by point 2, we obtain the equality in point 3. □

The notion of full comprehension is crucial to characterize when a given doctrine is the subobjects doctrine for a certain regular category. In particular, this characterisation allows to establish an equivalence between regular categories and doctrines.

Theorem 6.1. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. Then the following two conditions are equivalent:*

- *P has comprehensive diagonals, full comprehensions and satisfies (RUC);*
- *$\mathbf{C}$ is regular and $P \cong \mathsf{Sub}_{\mathbf{C}}$.*

In particular, $\mathbb{REG} \equiv \overline{\mathbb{EED}}_c$.

Proof. See [52, Prop. 5.3]. □

As a direct corollary of Theroem 6.1 we have the following equivalence:

Corollary 6.1. $\mathbb{BC} \equiv \overline{\mathbb{BHD}}_c$.

Proof. Recall that, by definition of boolean category (see [30, Sec. A1.4, p. 38]), we have that a regular category $\mathbf{C}$ is boolean if and only if the subobjects functor on $\mathbf{C}$ is a boolean hyperdoctrine. Hence, by Theorem 6.1 we have the equivalence $\mathbb{BC} \equiv \overline{\mathbb{BHD}}_c$.
□

6.3. The equivalence $\mathbb{BC} \equiv \mathbb{FOB}_f$

We can now prove our key result, illustrating that the notion of tabulation and that of full comprehension happen to be equivalent when we consider the doctrine associated with a cartesian bicategory.

Theorem 6.2. *A cartesian bicategory $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ is functionally complete if and only if the elementary and existential doctrine $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ has full comprehensions.*

Proof. Let $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ be functionally complete, and let us consider the doctrine $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ and an element $r : X \to I$ of $\mathbf{C}[X, I]$. We claim that the tabulation $i : X_r \to X$ of r is the comprehension of r.
First, notice that

$$\begin{aligned}
\mathbf{C}[i, I](r) &= i \fatsemi r && \text{(Def. } \mathbf{C}[-, I]\text{)}\\
&= i \fatsemi i^\dagger \fatsemi !^\circ_{X_r} && (i \text{ tabulation of } r)\\
&= id^\circ_{X_r} \fatsemi !^\circ_{X_r} && (i \text{ tabulation of } r)\\
&= !^\circ_{X_r}\\
&= \top_{X_r} && \text{(Table 2)}
\end{aligned}$$

Now suppose that $f : Z \to X$ is a map such that $\top_Z = \mathbf{C}[f, I](r)$. Since $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ is an elementary and existential doctrine, as noted in Remark 4.1, $\mathbf{C}[f, I] : \mathbf{C}[Y, I] \to \mathbf{C}[X, I]$ has a left adjoint $\exists_f : \mathbf{C}[X, I] \to \mathbf{C}[Y, I]$. Thus, from $\top_Z \leq \mathbf{C}[f, I](r)$ one can derive

$$\exists_f(\top_Z) \leq r. \tag{32}$$

Now, by means of the characterisation of $\exists_f$ in $\mathbf{C}[-, I]$ in Table 2, it holds that

$$\begin{aligned} f^\dagger \fatsemi !^\circ_Z &= \exists_f(\top_Z) && \text{(Table 2)} \\ &\leq r && (32) \end{aligned}$$

Therefore, one can apply Lemma 6.1 to derive that there exists a unique h such that $f = h \fatsemi i$. This concludes the proof that if $(\mathbf{C}, \blacktriangleleft^\circ, \blacktriangleright^\circ)$ is functionally complete then $\mathbf{C}[-, I] : \mathsf{Map}(\mathbf{C})^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ has comprehensions.

Now we need to prove that comprehensions are full. So let $r, r' : X \to I$ be two arrows and let i and i' their tabulations. If i factors through i', namely $i = g \fatsemi i'$ for some map $g : X_r \to X_{r'}$ then, by Table 1, we have that

$$i^\dagger = (i')^\dagger \fatsemi g^\dagger. \tag{33}$$

Hence, we can conclude that

$$\begin{aligned} r &= i^\dagger \fatsemi !^\circ_{X_r} && (i \text{ tabulation of } r) \\ &= (i')^\dagger \fatsemi g^\dagger \fatsemi !^\circ_{X_r} && (33) \\ &\leq (i')^\dagger \fatsemi !^\circ_{X_{r'}} && (!^\circ\text{-nat}) \\ &= r' && (i' \text{ tabulation of } r') \end{aligned}$$

Now we show that full comprehensions implies functionally completeness. So, let us consider an arrow $r : X \to I$. We claim that the comprehension $\{r\} : X_r \to X$ is a tabulation of r. Our proof crucially exploits Lemma 6.2. By the first point of Lemma 6.2, it holds that

$$\begin{aligned} \{r\}^\dagger \fatsemi !^\circ_{X_r} &= \exists_{\{r\}}(\top_{X_r}) && \text{(Table 2)} \\ &= r && \text{(Lemma 6.2.1)} \end{aligned}$$

By the third point of Lemma 6.2, it holds that

$$\begin{aligned} (\{r\} \otimes \{r\}) \fatsemi \blacktriangleright^\circ_{X_r} \fatsemi !^\circ_{X_r} &= \mathbf{C}[\{r\} \times \{r\}, I](\delta_X) && \text{(Table 2)} \\ &= \delta_{X_r} && \text{(Lemma 6.2.3)} \\ &= \blacktriangleright^\circ_{X_r} \fatsemi !^\circ_{X_r} && \text{(Table 2)} \end{aligned}$$

and therefore, by Lemma 2.1.4, it follows immediately that $\{r\} \fatsemi \{r\}^\dagger = id^\circ_{X_r}$. □

Combining Theorem 4.1 with Theorem 6.2 we obtain the following.

Corollary 6.2. *Let $P : \mathbf{C}^{\mathrm{op}} \longrightarrow \mathsf{InfSl}$ be an elementary and existential doctrine. Then the following two conditions are equivalent:*

- *P has comprehensive diagonals, full comprehension and satisfies (RUC);*
- *$\mathbf{C} \cong \mathsf{Map}(\mathsf{Rel}(P))$, $\mathsf{Rel}(P)$ functionally complete and $P \cong \mathsf{Rel}(P)[-, I]$.*

In particular, $\overline{\mathbb{EED}}_c \equiv \mathbb{CB}_f$.

Now, we can combine Corollary 6.2 with Theroem 6.1 obtaining as corollary the well-known equivalence (see [44] and [63]) between regular categories and functionally complete cartesian bicategories.

Corollary 6.3. $\mathbb{REG} \equiv \overline{\mathbb{EED}}_c \equiv \mathbb{CB}_f$.

Finally, by combining Corollary 6.2 with Corollary 6.1 and Theorem 5.2, we can easily prove the equivalence between $\mathbb{FOB}_f$ and $\mathbb{BC}$.

Corollary 6.4. $\mathbb{BC} \equiv \overline{\mathbb{BHD}}_c \equiv \mathbb{FOB}_f$.

Proof. By Theroem 5.2 we have the equivalences $\overline{\mathbb{BHD}} \equiv \mathbb{FOB}$, so we can combine this result with Corollary 6.2 obtaining the equivalence $\overline{\mathbb{BHD}}_c \equiv \mathbb{FOB}_f$. Therefore, combining this equivalence with Corollary 6.1 we obtain

$$\mathbb{BC} \equiv \overline{\mathbb{BHD}}_c \equiv \mathbb{FOB}_f.$$

□

7. Conclusions, related and future work

Theorems 5.1, 5.2 and Corollary 6.4 provide a solid bridge between functional and relational approaches to classical logic. The former rely on categorical structures that are usually defined by means of exactness properties; the latter on fo-bicategories which enjoy a purely equational presentation, much in the spirit of Boole's algebra and Peirce's calculus.

To obtain our results, we found it especially convenient to introduce the notion of *peircean bicategories*, which, by Theorem 3.1, offer a significantly more tractable characterisation of fo-bicategories.

The isomorphism between fo-bicategories and peircean bicategories may also be instrumental in establishing connections with *allegories* [9], arguably the most influential framework for categorical relational algebra. Since cartesian bicategories are equivalent to unitary pretabular allegories [64], we anticipate that those allegories in which homsets additionally form boolean algebras and where negation satisfies $(\neg\mathcal{M})$ are equivalent to fo-bicategories. Interestingly, property $(\neg\mathcal{M})$ holds in any Peirce allegory, as shown in Proposition 4.6.1 of [65], and in any heterogeneous relation algebra, as shown in Theorem 3.2.9 of [66]. Heterogeneous relation algebras are similar to peircean bicategories in that they are categories whose homsets carry a boolean algebra structure. However, instead of $(\neg\mathcal{M})$ they assume other implicational axioms such as Schröder's equivalences and Tarski's rule.

Boolean hyperdoctrines are used in [67] as a categorical treatment of another work of Peirce: *existential graphs* [68]. While the latter share some similarities with the graphical language of fo-bicategories there is one notable difference: negation is a primitive operator rather than a derived one, as it happens for instance also in [69] and Definition 3.1. In [43] and in § 3, it is emphasised how this choice makes the resulting calculus less algebraic in flavour, having to deal with convoluted rules such as the one for (de)iteration or properties which are not purely equational, such as $(\neg\mathcal{M})$.

Inspired by [67], another graphical language proposed in [70] and akin to Peirce's graphs is based on a decomposition of a hyperdoctrine into a bifibration. The categorical treatment there centers around the notion of monoidal *chiralities* [71], which are more closely aligned with the spirit of fo-bicategories. We believe our results may lay the groundwork for a deeper connection between fo-bicategories and chiralities.

A recent study [72] offers a relational interpretation of doctrines, although it pertains only to the regular fragment of first-order logic. It would be intriguing to explore the role of the additional "black structure" of first-order bicategories in this context. Closely related research–such as [73,74] and the references therein–focuses on categorical approaches to classical proof theory, often involving suitable variants of poset-enriched categories.

As future work, we aim to investigate how our characterisations might be extended to higher-order classical logic, which is categorically represented through the notion of *tripos* [61,75]. We believe the constructions and results developed here, together with the notion of tripos, could guide the definition of a variant of fo-bicategories–ideally, a purely equational one–capable of capturing higher-order classical logic.

An important result in database theory [76] demonstrates that the problem of query inclusion (i.e., entailment of regular-logic formulas) corresponds to the existence of morphisms between hypergraphs. This combinatorial insight received an elegant algebraic interpretation in [77] via cartesian bicategories. We hope that Theorem 3.1 may similarly lead to a combinatorial understanding of first-order logic.

CRediT authorship contribution statement

Filippo Bonchi: Writing – original draft, Writing – review & editing; **Alessandro Di Giorgio:** Writing – original draft, Writing – review & editing; **Davide Trotta:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Given his role as editor, Filippo Bonchi had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. - F.B. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Appendix to Section 3

Proof of Proposition 3.1. The four inclusions are equivalent since $\mathbf{C}[X, Y]$ is a boolean algebra. □

Proof of Proposition 3.2.

$$\begin{aligned} f \fatsemi c &= f \fatsemi \neg\neg c && \text{(Definition 3.1.1)} \\ &= \neg(f \fatsemi \neg c) && (\neg\mathcal{M}) \\ &= \neg f \mathbin{;} \neg\neg c && (10) \\ &= \neg f \mathbin{;} c. && \text{(Definition 3.1.1)} \end{aligned}$$

□

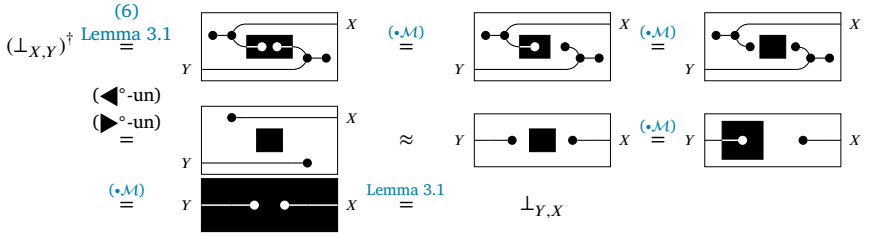
PROOF OF LEMMA 3.1.

$$
\begin{aligned}
c \vee d &= \neg(\neg c \wedge \neg d) && \text{(De Morgan)}\\
&= \neg(\blacktriangleleft^{\circ}_{X} \fatsemi (\neg c \otimes \neg d) \fatsemi \blacktriangleright^{\circ}_{Y}) && (5)\\
&= \neg(\neg\blacktriangleleft^{\bullet}_{X} \fatsemi \neg(c \otimes d) \fatsemi \neg\blacktriangleright^{\bullet}_{Y}) && (10)\\
&= \blacktriangleleft^{\bullet}_{X} \fatsemi (c \otimes d) \fatsemi \blacktriangleright^{\bullet}_{Y} && (10)
\end{aligned}
\qquad
\begin{aligned}
\bot &= \neg\top && \text{(De Morgan)}\\
&= \neg(!^{\circ}_{X} \fatsemi \mathrm{i}^{\circ}_{Y}) && (5)\\
&= \neg(\neg!^{\bullet}_{X} \fatsemi \neg\mathrm{i}^{\bullet}_{Y}) && (10)\\
&= !^{\bullet}_{X} \fatsemi \mathrm{i}^{\bullet}_{Y} && (10)
\end{aligned}
$$

□

Proposition A.1. *For any X, Y it holds that $(\bot_{X,Y})^{\dagger} = \bot_{Y,X}$.*

Proof.



□

PROOF OF LEMMA 3.2. We prove $(\neg c)^{\dagger} \leq \neg(c^{\dagger})$ below by means of Proposition 3.1.4.

$$
\begin{aligned}
(\neg c)^{\dagger} \wedge \neg\neg(c^{\dagger}) &= (\neg c)^{\dagger} \wedge c^{\dagger} && \text{(Definition 3.1.1)}\\
&= (\neg c \wedge c)^{\dagger} && \text{(Table 1)}\\
&= (\bot_{X,Y})^{\dagger} && \text{(Definition 3.1.1)}\\
&= \bot_{Y,X} && \text{(Proposition A.1)}
\end{aligned}
$$

The other inclusion, namely $\neg(c^{\dagger}) \leq (\neg c)^{\dagger}$, is proven below using the properties of $(\cdot)^{\dagger}$ in Table 1 and the inclusion proved above.

$$
\begin{aligned}
\neg(c^{\dagger}) &= (\neg(c^{\dagger}))^{\dagger^{\dagger}} && \text{(Table 1)}\\
&\leq (\neg(c^{\dagger^{\dagger}}))^{\dagger} && ((\neg c)^{\dagger} \leq \neg(c^{\dagger}))\\
&= (\neg c)^{\dagger} && \text{(Table 1)}
\end{aligned}
$$

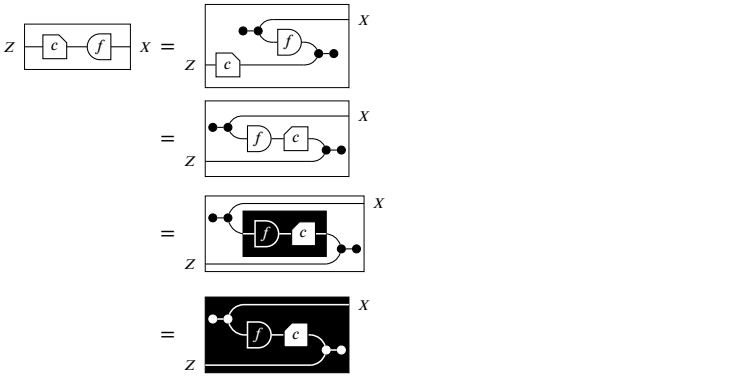
□

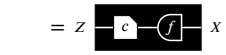
PROOF OF LEMMA 3.3.

$$
\begin{aligned}
c^{\dagger} &= (\neg\neg c)^{\dagger} && \text{(Definition 3.1.1)}\\
&= \neg\big((\neg c)^{\dagger}\big) && \text{(Lemma 3.2)}\\
&= (\neg\neg c)^{\ddagger} && (9)\\
&= c^{\ddagger} && \text{(Definition 3.1.1)}
\end{aligned}
$$

□

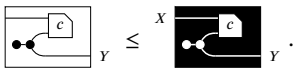
PROOF OF LEMMA 3.4. We give a proof for the second equality only. For the first one, the proof follows an argument analogous to that of Proposition 3.2.



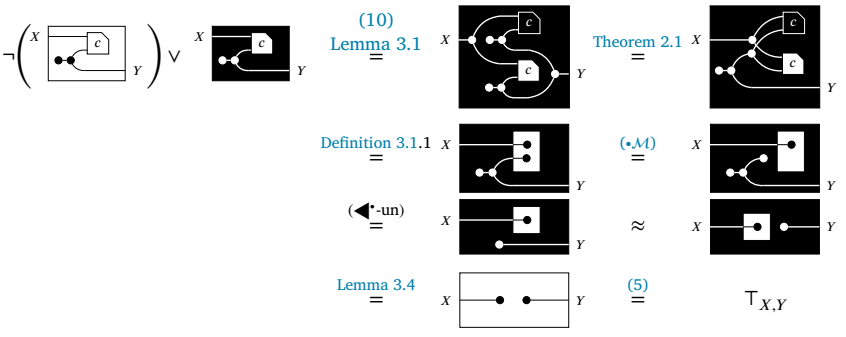


(9)

□

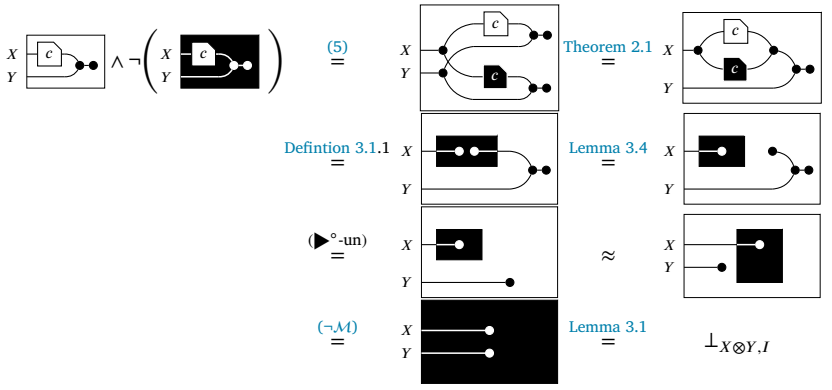
Proposition A.2. *For any* $c : X \otimes Y \to I$*, it holds that* 

Proof. Observe that the following holds:

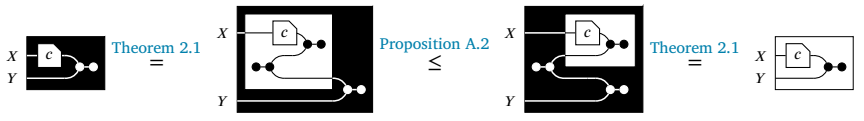


and thus we conclude by Proposition 3.1.3. □

PROOF OF LEMMA 3.5. We prove the left-to-right inclusion below, by means of Proposition 3.1.4.



For the other inclusion observe that the following holds.



□

References

[1] C.S. Peirce, The logic of relatives, Monist 7 (2) (1897) 161–217. http://www.jstor.org/stable/27897407.
[2] G. Frege, Begriffsschrift und andere Aufsätze, Georg Olms Verlag, 1977.
[3] W. Ewald, The emergence of first-order logic, The Stanford Encyclopedia of Philosophy, Winter 2018, E.N. Zalta (Ed.), Metaphysics Research Lab, Stanford University (2018). https://plato.stanford.edu/archives/win2018/entries/logic-firstorder-emergence/.
[4] A. De Morgan, On the syllogism, no. IV. and on the logic of relations, Transactions of the Cambridge Philosophical Society 10 (1860) 173–230.
[5] G. Boole, The Mathematical Analysis of Logic, Philosophical Library, 1847.
[6] D. Monk, On representable relation algebras, Mich. Math. J. 11 (3) (1964) 207 – 210. https://doi.org/10.1307/mmj/1028999131
[7] I. Hodkinson, S. Mikulás, Axiomatizability of reducts of algebras of relations, Algebra Univers. 43 (2) (2000) 127–156. https://doi.org/10.1007/s000120050150
[8] V.N. Redko, On defining relations for the algebra of regular events, Ukr. Mat. Zh. 16 (1964) 120–126.
[9] P. Freyd, A. Scedrov, Categories, Allegories, Vol. 39 of *North-Holland Mathematical Library*, Elsevier B.V, 1990.
[10] B. Bollig, A. Finkel, A. Suresh, Bounded reachability problems are decidable in FIFO machines, in: I. Konnov, L. Kovács (Eds.), 31st International Conference on Concurrency Theory (CONCUR 2020), vol. 171 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2020, pp. 49:1–49:17. https://doi.org/10.4230/LIPIcs.CONCUR.2020.49
[11] D. Pous, On the positive calculus of relations with transitive closure, in: R. Niedermeier, B. Vallée (Eds.), 35th Symposium on Theoretical Aspects of Computer Science, STACS 2018, February 28, to March 3, 2018, Caen, France, vol. 96 of *LIPIcs*, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2018, pp. 3:1–3:16. https://doi.org/10.4230/LIPIcs.STACS.2018.3
[12] E.F. Codd, A relational model of data for large shared data banks, Commun. ACM 26 (1) (1983) 64–69.
[13] V.R. Pratt, Semantical considerations on Floyd-Hoare logic, in: 17th Annual Symposium on Foundations of Computer Science (Sfcs 1976), IEEE, 1976, pp. 109–121.

[14] C. Hoare, H. Jifeng, The weakest prespecification, Part I, Fundam. Inform. 9 (1) (1986) 51–84.
[15] S.B. Lassen, Relational reasoning about contexts, High. Order Oper. Tech. Semant. 91 (1998).
[16] R. Bird, O. De Moor, The algebra of programming, NATO ASI DPD 152 (1996) 167–203.
[17] U.D. Lago, F. Gavazzo, A relational theory of effects and coeffects, Proc. ACM Program. Lang. 6 (POPL) (2022) 1–28. https://doi.org/10.1145/3498692
[18] D. Pous, Kleene algebra with tests and Coq tools for while programs, in: Interactive Theorem Proving: 4th International Conference, ITP 2013, Rennes, France, July 22–26, 2013. Proceedings 4, Springer, 2013, pp. 180–196.
[19] D. Pous, Automata for Relation Algebra and Formal Proofs, Ph.D. thesis , ENS Lyon, 2016.
[20] A. Krauss, T. Nipkow, Proof pearl: regular expression equivalence and relation algebra, J. Autom. Reason. 49 (1) (2012) 95–106. https://doi.org/10.1007/s10817-011-9223-4
[21] L. Löwenheim, Über möglichkeiten im relativkalkül, Math. Ann. 76 (4) (1915) 447–470.
[22] A. Tarski, On the calculus of relations, J. Symb. Log. 6 (3) (1941) 73–89. https://doi.org/10.2307/2268577
[23] C.J. Everett, S. Ulam, Projective algebra I, Am. J. Math. 68 (1) (1946) 77–88.
[24] L. Henkin, Cylindric Algebras, Part I, J.D. Monk, A. Tarski, Studies in Logic and the Foundations of Mathematics, 64, North-Holland Publishing Company, Amsterdam (1971).
[25] W.V. Quine, Predicate-functor logics, in: J.E. Fenstad (Ed.), Proceedings of the Second Scandinavian Logic Symposium, vol. 63 of *Studies in Logic and the Foundations of Mathematics*, Elsevier, 1971, pp. 309–315. https://doi.org/10.1016/S0049-237X(08)70850-4
[26] F.W. Lawvere, Adjointness in foundations, Dialectica 23 (1969) 281–296.
[27] F.W. Lawvere, Diagonal arguments and cartesian closed categories, in: Category Theory, Homology Theory and their Applications, vol. 2, Springer, 1969, pp. 134–145.
[28] F.W. Lawvere, Equality in hyperdoctrines and comprehension schema as an adjoint functor, in: A. Heller (Ed.), New York Symposium on Application of Categorical Algebra, vol. 2, American Mathematical Society, 1970, pp. 1–14.
[29] P.T. Johnstone, Topos Theory, Academic Press, 1977.
[30] P.T. Johnstone, Sketches of an Elephant: A Topos Theory Compendium, Vol. 2 of *Studies in Logic and the foundations of mathematics*, Oxford Univ. Press, 2002.
[31] A. Joyal, R. Street, The geometry of tensor calculus, I, Adv. Math. 88 (1) (1991) 55–112. https://doi.org/10.1016/0001-8708(91)90003-P
[32] P. Selinger, A survey of graphical languages for monoidal categories, in: B. Coecke (Ed.), New Structures for Physics, vol. 813 of *Lecture Notes in Physics*, Springer, Berlin, Heidelberg, 2010, pp. 289–355. https://doi.org/10.1007/978-3-642-12821-9_4
[33] D. Stein, S. Staton, Probabilistic programming with exact conditions, J. ACM (2023).
[34] F. Bonchi, F. Gadducci, A. Kissinger, P. Sobocinski, F. Zanasi, String diagram rewrite theory i: rewriting with frobenius structure, J. ACM 69 (2) (2022) 14:1–14:58. https://doi.org/10.1145/3502719
[35] F. Bonchi, J. Holland, R. Piedeleu, P. Sobociński, F. Zanasi, Diagrammatic algebra: from linear to concurrent systems, Proc. ACM Program. Lang. 3 (POPL) (2019) 25:1–25:28. https://doi.org/10.1145/3290338
[36] F. Bonchi, P. Sobocinski, F. Zanasi, Full abstraction for signal flow graphs, in: Proceedings of the 42nd Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, POPL '15, Association for Computing Machinery, New York, NY, USA, 2015, pp. 515–526. https://doi.org/10.1145/2676726.2676993
[37] B. Coecke, R. Duncan, Interacting quantum observables: categorical algebra and diagrammatics, New J. Phys. 13 (4) (2011) 043016. https://doi.org/10.1088/1367-2630/13/4/043016
[38] B. Fong, P. Sobociński, P. Rapisarda, A categorical approach to open and interconnected dynamical systems, in: Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '16, Association for Computing Machinery, New York, NY, USA, 2016, pp. 495–504. https://doi.org/10.1145/2933575.2934556
[39] B. Fong, D. Spivak, String diagrams for regular logic (extended abstract), in: J. Baez, B. Coecke (Eds.), Applied Category Theory 2019, vol. 323 of *Electronic Proceedings in Theoretical Computer Science*, Open Publishing Association, 2020, pp. 196–229. https://doi.org/10.4204/eptcs.323.14
[40] D.R. Ghica, A. Jung, Categorical semantics of digital circuits, in: 2016 Formal Methods in Computer-Aided Design (FMCAD), 2016, pp. 41–48. https://doi.org/10.1109/FMCAD.2016.7886659
[41] K. Muroya, S.W.T. Cheung, D.R. Ghica, The geometry of computation-graph abstraction, in: A. Dawar, E. Grädel (Eds.), Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2018, Oxford, UK, July 09–12, 2018, ACM, 2018, pp. 749–758. https://doi.org/10.1145/3209108.3209127
[42] R. Piedeleu, F. Zanasi, A string diagrammatic axiomatisation of finite-state automata, in: S. Kiefer, C. Tasson (Eds.), Foundations of Software Science and Computation Structures, Lecture Notes in Computer Science, Springer International Publishing, Cham, 2021, pp. 469–489. https://doi.org/10.1007/978-3-030-71995-1_24
[43] F. Bonchi, A. Di Giorgio, N. Haydon, P. Sobocinski, Diagrammatic algebra of first order logic, in: Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '24, Association for Computing Machinery, New York, NY, USA, 2024. https://doi.org/10.1145/3661814.3662078
[44] A. Carboni, R.F.C. Walters, Cartesian bicategories I, J. Pure Appl. Algebra 49 (1) (1987) 11–32. https://doi.org/10.1016/0022-4049(87)90121-6
[45] J.R.B. Cockett, J. Koslowski, R.A.G. Seely, Introduction to linear bicategories, Math. Struct. Comput. Sci. 10 (2) (2000) 165–203.
[46] F. Bonchi, A. Santamaria, J. Seeber, P. Sobociński, On doctrines and cartesian bicategories, in: 9th Conference on Algebra and Coalgebra in Computer Science (CALCO 2021), Schloss Dagstuhl-Leibniz-Zentrum für Informatik, 2021.
[47] M.E. Maietti, G. Rosolini, Quotient completion for the foundation of constructive mathematics, Log. Univers. 7 (3) (2013) 371–402.
[48] M.E. Maietti, G. Rosolini, Elementary quotient completion, Theory App. Categ. 27 (17) (2013) 445–463.
[49] M.E. Maietti, G. Rosolini, Unifying exact completions, Appl. Categ. Structures 23 (2013) 43–52.
[50] T. Fox, Coalgebras and cartesian categories, Commun. Algebra 4 (7) (1976) 665–667. https://doi.org/10.1080/00927877608822127
[51] B. Jacobs, Categorical Logic and Type Theory, Vol. 141 of *Studies in Logic and the foundations of mathematics*, North Holland Publishing Company, 1999.
[52] M.E. Maietti, F. Pasquali, G. Rosolini, Triposes, exact completions, and Hilbert's ε-operator, Tbil. Math. J. 10 (3) (2017) 141–166.
[53] F. Bonchi, A. Di Giorgio, D. Trotta, When lawvere meets peirce: an equational presentation of boolean hyperdoctrines, in: R. Královic, A. Kucera (Eds.), 49th International Symposium on Mathematical Foundations of Computer Science, MFCS 2024, August 26–30, 2024, Bratislava, Slovakia, vol. 306 of *LIPIcs*, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024, pp. 30:1–30:19. https://doi.org/10.4230/LIPICS.MFCS.2024.30
[54] C.S. Peirce, Studies in logic. By Members of the Johns Hopkins University, Little, Brown, and Company, 1883.
[55] A. Di Giorgio, Diagrammatic Algebras of Relations, Ph.D. thesis, University of Pisa, 2024.
[56] S. Lack, Composing PROPs, Theory Appl. Categ. 13 (9) (2004) 147–163. http://www.tac.mta.ca/tac/volumes/13/9/13-09abs.html.
[57] B. Coecke, R. Duncan, Interacting quantum observables, in: L. Aceto, I. Damgård, L.A. Goldberg, M.M. Halldórsson, A. Ingólfsdóttir, I. Walukiewicz (Eds.), Automata, Languages and Programming, Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2008, pp. 298–310. https://doi.org/10.1007/978-3-540-70583-3_25
[58] F. Bonchi, D. Pavlovic, P. Sobocinski, Functorial semantics for relational theories, (2017). arXiv preprint arXiv:1711.08699
[59] F.W. Lawvere, Functorial Semantics of Algebraic Theories, Ph.D. thesis, Columbia University, New York, NY, USA, 1963.
[60] M.E. Maietti, D. Trotta, A characterization of generalized existential completions, Ann. Pure Appl. Log. 174 (4) (2023) 103234.
[61] J.M.E. Hyland, P.T. Johnstone, A.M. Pitts, Tripos theory, Math. Proc. Camb. Phil. Soc. 88 (1980) 205–232.
[62] M.E. Maietti, F. Pasquali, G. Rosolini, Quasitoposes as elementary quotient completions, 2024, arXiv:2111.15299 [math.LO]
[63] B. Fong, D.I. Spivak, Regular and relational categories: revisiting 'cartesian bicategories I', arXiv (2019). https://api.semanticscholar.org/CorpusID:202540886.
[64] P.M.W. Knijnenburg, F. Nordemann, Two Categories of Relations, Citeseer, 1994.
[65] J.-P. Olivier, D. Serrato, Peirce allegories. Identities involving transitive elements and symmetrical ones, J. Pure Appl. Algebra 116 (1–3) (1997) 249–271.
[66] G. Schmidt, C. Hattensperger, M. Winter, Heterogeneous relation algebra, in: Relational Methods in Computer Science, Springer, 1997, pp. 39–53.
[67] G. Brady, T. Trimble, A categorical interpretation of C.S. Peirce's propositional logic Alpha, J. Pure Appl. Alg. 149 (2000) 213–239. https://doi.org/10.1016/S0022-4049(98)00179-0

[68] D.D. Roberts, The Existential Graphs of Charles S. Peirce, De Gruyter Mouton, 1973.
[69] N. Haydon, P. Sobociński, Compositional diagrammatic first-order logic, in: 11th International Conference on the Theory and Application of Diagrams (DIAGRAMS 2020), 2020.
[70] P.-A. Melliès, N. Zeilberger, A bifibrational reconstruction of Lawvere's presheaf hyperdoctrine, in: Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science, 2016, pp. 555–564.
[71] P.-A. Melliès, Dialogue categories and chiralities, Publ. Res. Inst. Math. Sci. 52 (4) (2016) 359–412.
[72] F. Dagnino, F. Pasquali, Quotients and extensionality in relational doctrines, in: M. Gaboardi, F. van Raamsdonk (Eds.), 8th International Conference on Formal Structures for Computation and Deduction (FSCD 2023), vol. 260 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2023, pp. 25:1–25:23. https://doi.org/10.4230/LIPIcs.FSCD.2023.25
[73] M. Hyland, Abstract interpretation of proofs: classical propositional calculus, in: J. Marcinkowski, A. Tarlecki (Eds.), Computer Science Logic, Springer Berlin Heidelberg, Berlin, Heidelberg, 2004, pp. 6–21.
[74] C. Führmann, D. Pym, On categorical models of classical logic and the geometry of interaction, Math. Struct. Comput. Sci. 17 (5) (2007) 957–1027. https://doi.org/10.1017/S0960129507006287
[75] A.M. Pitts, Tripos theory in retrospect, Math. Struct. Comp. Science 12 (2002) 265–279.
[76] A.K. Chandra, P.M. Merlin, Optimal implementation of conjunctive queries in relational data bases, in: Proceedings of the Ninth Annual ACM Symposium on Theory of Computing, STOC '77, Association for Computing Machinery, New York, NY, USA, 1977, pp. 77–90. https://doi.org/10.1145/800105.803397
[77] F. Bonchi, J. Seeber, P. Sobocinski, Graphical conjunctive queries, in: D. Ghica, A. Jung (Eds.), 27th EACSL Annual Conference on Computer Science Logic (CSL 2018), vol. 119 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, Dagstuhl, Germany, 2018, pp. 13:1–13:23. https://doi.org/10.4230/LIPIcs.CSL.2018.13